KronHiPPO-STGP: Online Bayesian Spatiotemporal Modeling with Evolving Memory

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Abstract

Spatiotemporal modeling often requires capturing both global trends and correlated uncertainty across space and time. In addition, since data arrive sequentially in typical applications, online modeling and prediction are required. We introduce KronHiPPO-STGP, an online Bayesian spatiotemporal model that jointly updates a parametric global trend (captured by linear regression) and a residual model by a Gaussian process (GP) with a separable spatiotemporal kernel. To maintain fixed memory usage during online inference, we use Online HiPPO Sparse Variational GP (Online HiPPO-SVGP) along the temporal axis, which updates the interdomain inducing variables by evolving an ODE-based recurrent memory. To address computational speed challenges, we solve the ODE evolution via Bessel function computations, and we leverage the Kronecker structure of the residual GP to avoid dense matrix inversion. These have reduced the computational complexity from $\mathcal{O}(M^6)$ to $\mathcal{O}(M^3)$, where M is the number of inducing variables along each spatial and temporal axis. Experiments on climate, public health and traffic modeling datasets show that KronHiPPO-STGP maintains memory costs independent of stream length and achieves accurate predictions with uncertainty quantification in online regression settings.

1 INTRODUCTION

Spatiotemporal data arise in real-world sequential records, such as climate observations (Muñoz-Sabater et al., 2021), public health monitoring data (Silk et al., 2023), and traffic records (Li et al., 2018). To model these data, a model needs to capture not only overall trends but also spatially and temporally correlated uncertainty. A standard approach to modeling spatiotemporal data is to combine a mean function, typically linear regression using spatial and temporal features, with a residual model, such as a Gaussian process (GP) with a separable kernel comprising spatial and temporal components (Cressie, 1993; Stein, 1999; Rasmussen and Williams, 2005; Flaxman et al., 2015). This decomposition provides a natural framework for capturing both global mean trends and correlated uncertainty across space and time.

In real-world applications, data arrive sequentially and the temporal horizon expands, so we need to update predictions in an online manner. However, repeatedly fitting the model, especially the residual GP part, using all accumulated data at each update requires a computational cost that grows with the temporal horizon (Bui et al., 2017). Therefore, a compressed representation is required to summarize the data streams (Csató and Opper, 2002; Titsias, 2009; Maddox et al., 2021).

To address this challenge, we construct temporal inducing variables for the residual GP based on Online HiPPO Sparse Variational GPs (Online HiPPO-SVGPs, OHSVGP; Chen et al., 2025), which use interdomain inducing variables (Lázaro-Gredilla and Figueiras-Vidal, 2009) defined on an orthogonal polynomial basis, with the projection given by solving the HiPPO ODE (Gu et al., 2020). This approach allows us to maintain a fixed-size representation of the temporal history, which is then updated by a joint online Bayesian inference method for the joint posterior of the linear trend and the residual GP.

Two further computational difficulties arise in this online setting.

- First, HiPPO-SVGP requires numerical ODE integration to compute the covariance matrices of the temporal inducing variables. We address this by

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using a spherical Bessel function identity (Watson, 1944, § 3.32) to compute the required covariance matrices analytically, avoiding numerical integration of the HiPPO ODE.

- Second, the standard approach to online update the inducing variables involves $O((M_s M_t)^3)$ computation, where M_s and M_t are the number of spatial inducing points and temporal inducing variables, respectively. Similarly to the work by Rakitsch et al. (2013), we address this by exploiting the Kronecker structure of kernel matrices in the prior and the likelihood to reduce the computational cost of posterior inference from $\mathcal{O}((M_s M_t)^3)$ to $\mathcal{O}(M_s^3 + M_t^3)$.

To summarize, we propose KronHiPPO-STGP, a computationally and memory efficient online Bayesian method that jointly updates a parametric trend and a GP residual with separable spatiotemporal kernel. Our contributions include

(1) A joint online variational inference method to update the joint posterior of the linear trend's parameters and the residual GP function; and

(2) Computational methods as aforementioned to make the proposed method scalable.

We evaluate KronHiPPO-STGP on real-world datasets from three domains: climate, public health, and traffic. We show that KronHiPPO-STGP improves predictive accuracy and uncertainty calibration compared to existing methods, while keeping the memory cost independent of the stream length.

2 BACKGROUND

Section 2.1 introduces a linear trend with a spatiotemporal GP residual, which forms the basis of our model. We then review sparse variational GPs (SVGPs) and interdomain GPs in Section 2.2, and then describe Online HiPPO-SVGP, which we use for temporal modeling, in Section 2.3.

2.1 Linear trends with spatiotemporal GP residual

In spatiotemporal statistics, it is common to model the observation as a global linear trend with spatiotemporal GP residual, which is also known as *universal kriging* (Cressie, 1993). The model is given by

$$y(t,s) = \underbrace{\phi(t,s)^\mathsf{T}\beta}_{\text{linear trend}} + \underbrace{f(t,s)}_{\text{GP residual}} + \epsilon, \tag{1}$$

where $\phi(t,s) \in \mathbb{R}^{d_\beta}$ is a predefined basis function, $\beta \in \mathbb{R}^{d_\beta}$ is the coefficients, $f(t,s)$ is a GP with covariance function $k((t,s),(t',s'))$, and ϵ is the Gaussian noise with variance σ_y^2. the GP residual captures spatiotemporally correlated uncertainty in the data which is not captured by the linear trends.

In our work, we define a Gaussian prior for the linear trend coefficient as $\beta \sim \mathcal{N}(m_{\beta,0}, S_{\beta,0})$. For the residual GP, we adopt a separable spatiotemporal kernel to enable efficient computation of the GP posterior:

$$k\big((t,s),(t',s')\big) = k_t(t,t')\, k_s(s,s'). \tag{2}$$

When observations are given at the same spatial locations at every time point, a separable kernel gives a Kronecker-structured covariance matrix. We make this assumption in our work and use this structure to efficiently compute the joint posterior of the trend coefficients and the GP inducing variables.

2.2 Sparse variational Gaussian processes and interdomain Gaussian processes

The computational cost of exact GP posterior inference scales as $\mathcal{O}(N^3)$ with the number of observations N. Even with separable spatiotemporal kernels, the cost is still $\mathcal{O}(N_s^3 N_t^3)$ for a grid of N_s locations and N_t time-points, which is prohibitive for large datasets. To address this, sparse variational Gaussian processes (SVGPs; Titsias, 2009; Hensman et al., 2013) introduce a finite set of inducing variables that summarize information from the data. Given M inducing points in the input space $\widetilde{x}_1, \ldots, \widetilde{x}_M$, define the inducing variables $u = (f(\widetilde{x}_1), \ldots, f(\widetilde{x}_M))^\mathsf{T} \in \mathbb{R}^M$ in the output space. Under a zero-mean GP prior, $p(u) = \mathcal{N}(0, K_{uu})$, where element of $K_{uu} \in \mathbb{R}^{M\times M}$ is given by $[K_{uu}]_{ij} = k(\widetilde{x}_i, \widetilde{x}_j)$. The approximate posterior is constructed as

$$q(f) = \int p(f \mid u) q(u)\, du, \tag{3}$$

$$q(u) = \mathcal{N}(m_u, S_u). \tag{4}$$

Here, m_u and S_u are the variational mean and covariance, and they are optimized to minimize the variational free energy (VFE; Titsias, 2009; Hensman et al., 2013), which is equivalent to maximizing the evidence lower bound (ELBO) of the marginal likelihood. The VFE objective is given by

$$\mathcal{F} = -\mathbb{E}_{q(f)}[\log p(y \mid f)] + \mathrm{KL}[q(u) \,\|\, p(u)]\,. \tag{5}$$

This variational formulation also allows for interdomain inducing variables (Lázaro-Gredilla and Figueiras-Vidal, 2009), which are defined as linear functionals of the GP,

$$u_m = \int f(x)\psi_m(x)\, dx, \tag{6}$$

where ψ_m is a predefined basis function. The standard SVGP corresponds to ψ_m being a Dirac delta function located at $\widetilde{x}_m$. When ψ_m has broad support, compared to standard inducing variables, which are localized to specific input locations, interdomain inducing variables can capture more global information and are particularly useful for modeling long-range dependencies in sequential data.

2.3 Online HiPPO-SVGP

HiPPO inducing variables. Online HiPPO-SVGP (Chen et al., 2025) utilizes interdomain inducing variables defined on a specific orthogonal polynomial basis, with the projection given by solving the HiPPO ODE (Gu et al., 2020). At time t, HiPPO compresses the history $\{f(\tau) : 0 \le \tau \le t\}$ into a fixed-dimensional state. Let us rewrite Eq. (6) to make the weighting measure $\omega^{(t)}(x)$ explicit:

$$u_m^{(t)} = \int f(x) g_m^{(t)}(x) \omega^{(t)}(x)\, dx. \tag{7}$$

Instead of explicitly computing the inner products between f and basis functions $g_m^{(t)}$ under measure $\omega^{(t)}(x)$ in Eq. (7), the projections can be computed by solving the following ODE:

$$\frac{d}{dt}u(t) = A_{\mathrm{H}}(t)u(t) + b_{\mathrm{H}}(t)f(t), \tag{8}$$

where $u(t) \in \mathbb{R}^M$ is the projection coefficients, and $A_{\mathrm{H}}(t) \in \mathbb{R}^{M\times M}$ and $b_{\mathrm{H}}(t) \in \mathbb{R}^M$ are determined by the choice of basis and weighting measure. Chen et al. (2025) adopt HiPPO-LegS, which uses scaled Legendre polynomials as the basis and a uniform measure on $[0, t]$, which means $\omega^{(t)}(x) = 1/t$ for $x \in [0, t]$. Writing $\psi_m^{(t)}(x) = g_m^{(t)}(x)\omega^{(t)}(x)$, the covariance matrices for input points $X = (x_1, \dots, x_N)$ are

$$[K_{uf}^{(t)}]_{mi} = \int_0^t \psi_m^{(t)}(\tau) k(\tau, x_i)\, d\tau, \tag{9}$$

$$[K_{uu}^{(t)}]_{mn} = \int_0^t \int_0^t \psi_m^{(t)}(\tau) k(\tau, \tau') \psi_n^{(t)}(\tau')\, d\tau\, d\tau'. \tag{10}$$

Covariance computation with random Fourier features (RFF). Here we describe how to compute the covariance matrices $K_{uf}^{(t)}$ and $K_{uu}^{(t)}$ in Eqs. (9) and (10). By Bochner's theorem, the normalized Fourier transform of the stationary kernel gives the probability density over frequencies (Rahimi and Recht, 2007):

$$p(\lambda) = \frac{1}{2\pi\sigma_f^2} \int_{\mathbb{R}} k(h) e^{-\mathrm{i}\lambda h}\, dh, \tag{11}$$

where $k(h) := k(\tau, \tau + h)$ is the kernel as a function of the time difference h, and $\sigma_f^2 = k(0)$ is the kernel variance. Now we see that the kernel can be expressed as an expectation over p:

$$\begin{aligned} k(\tau, \tau') &= \sigma_f^2 \mathbb{E}_{\lambda\sim p}\big[\cos\big(\lambda(\tau - \tau')\big)\big] \\ &\approx \frac{2\sigma_f^2}{D} \sum_{j=1}^{D/2} \cos\big(\lambda_j(\tau - \tau')\big), \end{aligned} \tag{12}$$

where D is an even number of Fourier features and $\lambda_1, \dots, \lambda_{D/2} \sim p$ are sampled frequencies. By changing the form, we can express the kernel as an inner product of features $k(\tau, \tau') \approx z(\tau)^{\mathsf{T}} z(\tau')$, where:

$$z(\tau) = \sqrt{\frac{2\sigma_f^2}{D}} \begin{bmatrix} (\sin(\lambda_j\tau))_{j=1}^{D/2} \\ (\cos(\lambda_j\tau))_{j=1}^{D/2} \end{bmatrix}. \tag{13}$$

To improve numerical stability, Chen et al. (2025) use this approximation with shared features to compute both K_{uu} and K_{uf} in their implementation. Substituting the approximation into Eq. (10) separates the double integral into products of single integrals:

$$[Z(t)]_{mj} = \int_0^t \psi_m^{(t)}(\tau) z_j(\tau)\, d\tau, \tag{14}$$

$$K_{uu}^{(t)} \approx Z(t) Z(t)^{\mathsf{T}}. \tag{15}$$

Each column of $Z(t) \in \mathbb{R}^{M\times D}$ is a HiPPO projection of one Fourier feature, so the columns can be updated independently by numerically solving

$$\frac{d}{dt}Z(t) = A_{\mathrm{H}}(t)Z(t) + b_{\mathrm{H}}(t) z(t)^{\mathsf{T}}. \tag{16}$$

Similarly, for fixed input locations X, the cross-covariance satisfies

$$\frac{d}{dt}K_{uf}^{(t)} = A_{\mathrm{H}}(t)K_{uf}^{(t)} + b_{\mathrm{H}}(t) k(t, X), \tag{17}$$

where $k(t, X) = [k(t, x_1), \dots, k(t, x_N)]$.

When a new batch arrives, the approximate posterior is updated without access to previous observations (Bui et al., 2017; Chen et al., 2025). We adapt this approach in our setting and describe the details in the next section.

3 METHOD

We consider the online setting: data arrives sequentially in batches $\mathcal{D}_1, \dots, \mathcal{D}_K$, where each batch $\mathcal{D}_k = \{(x_{ki}, y_{ki})\}_{i=1}^{n_k}$ contains n_k observations with $x_{ki} = (t_{ki}, s_{ki})$, and the batches are ordered in time, i.e., $\max_i t_{k-1,i} \le \min_i t_{ki}$ for $k = 2, \dots, K$. Our first attempt considered a mean-field approximation to the posterior (i.e., $q(\beta, f) = q(\beta)q(f)$) and directly incorporated the online variational inference approach to

update the temporal interdomain inducing variables as in Chen et al. (2025). This approach failed, however, due to the mismatch between the historical residuals computed with staled β from history and that computed with the current β, so that the evolved recurrent memory states are no longer valid for the current interdomain inducing point projections. The mean-field approach cannot correct this error without access to historical data, which is not available in online setting. Instead we propose a joint online variational inference approach to update the parametric trend and the GP residual as follows, and introduce computational techniques to make the method scalable.

3.1 Online inference for the additive GP model

To facilitate joint posterior inference, we first define the joint state of the linear trend coefficient β and the collection of the residual GP's inducing variables u_k (M_s spatial and M_t temporal, respectively) at online inference step k

$$\xi_k = \begin{bmatrix} \beta \\ u_k \end{bmatrix} \in \mathbb{R}^{d_\beta + M_s M_t}, \beta \in \mathbb{R}^{d_\beta}, u_k \in \mathbb{R}^{M_s M_t}. \quad (18)$$

We assume a Gaussian prior on ξ_k

$$\begin{aligned} p_k(\xi_k) &= p(\beta) p_k(u_k) = \mathcal{N}(\xi_k; m_0, \Lambda_{0,k}^{-1}), \\ m_0 &= \begin{bmatrix} m_{\beta,0} \\ 0 \end{bmatrix}, \quad \Lambda_{0,k}^{-1} = \mathrm{diag}(S_{\beta,0}, K_{uu,k}). \end{aligned} \quad (19)$$

where $m_{\beta,0}$ and $S_{\beta,0}$ are the prior mean and covariance of the trend coefficients, and $K_{uu,k}$ is the prior covariance of the inducing variables at step k.

Suppose that we have already observed the first $k-1$ batches and have obtained the approximate posterior $q_{k-1}(\xi_{k-1})$ at step $k-1$. When the new batch $\mathcal{D}_k$ arrives, we update the approximate posterior to $q_k(\xi_k)$ by maximizing the online ELBO of Bui et al. (2017), extended to the joint state:

$$\begin{aligned} \mathcal{L}_k^{\text{joint}} = \sum_{i=1}^{n_k} & \mathbb{E}_{q_k(\beta, f_{ki})}[\log p(y_{ki} \mid \beta, f_{ki})] \\ & - \mathrm{KL}[q_k(\xi_k) \,\|\, p_k(\xi_k)] \\ & + \mathrm{KL}[\widetilde{q}_k(\xi_{k-1}) \,\|\, p_{k-1}(\xi_{k-1})] \\ & - \mathrm{KL}[\widetilde{q}_k(\xi_{k-1}) \,\|\, q_{k-1}(\xi_{k-1})]. \end{aligned} \quad (20)$$

where $f_{ki} = f(x_{ki})$ and

$$\begin{aligned} q_k(\xi_k) &= \mathcal{N}(\xi_k; m_k, S_k), \\ q_k(\beta, f_{ki}) &= \int p_k(f_{ki} \mid u_k) q_k(\beta, u_k)\, du_k. \end{aligned} \quad (21)$$

The distribution $\widetilde{q}_k(\xi_{k-1})$ is obtained from $q_k(\xi_k)$ using the GP prior conditional $p_k(u_{k-1} \mid u_k)$:

$$\widetilde{q}_k(\xi_{k-1}) = \int p_k(u_{k-1} \mid u_k) q_k(\beta, u_k)\, du_k. \quad (22)$$

Applying the Gaussian update by Bui et al. (2017) to the joint state ξ_k gives the optimal update that maximizes the online ELBO in Eq. (20):

$$\begin{aligned} q_k^*(\xi_k) \propto\, & p_k(\xi_k) \\ & \times \exp\Bigg\{ \mathbb{E}_{p_k(f_k \mid u_k)}[\log p(y_k \mid \beta, f_k)] \\ & + \mathbb{E}_{p_k(u_{k-1} \mid u_k)}\left[\log \frac{q_{k-1}(\xi_{k-1})}{p_{k-1}(\xi_{k-1})}\right] \Bigg\}. \end{aligned} \quad (23)$$

Here, $y_k = (y_{k1}, \ldots, y_{kn_k})^\mathsf{T}$ and $f_k = (f_{k1}, \ldots, f_{kn_k})^\mathsf{T}$ collect the observations and corresponding GP values in $\mathcal{D}_k$. The first term in the exponent of Eq. (23) accounts for the new batch $\mathcal{D}_k$, and the second term accounts for the previous batches $\mathcal{D}_1, \ldots, \mathcal{D}_{k-1}$.

With Gaussian likelihood, $q_k^*(\xi_k)$ is also Gaussian: with $A_k := K_{f_k u_k} K_{uu,k}^{-1}$, where K_{ab} is the prior covariance between a and b, $J_k := [\Phi_k \; A_k]$, $[\Phi_k]_{i,:} := \phi(x_{ki})^\mathsf{T}$,

$$\begin{aligned} q_k^*(\xi_k) &= \mathcal{N}(\xi_k; m_k, S_k), \\ S_k^{-1} &= \Lambda_{0,k} + \sigma_y^{-2} J_k^\mathsf{T} J_k + M_{k-1,k}^\mathsf{T} R_{k-1} M_{k-1,k}, \\ m_k &= S_k\big(\Lambda_{0,k} m_0 + \sigma_y^{-2} J_k^\mathsf{T} y_k + M_{k-1,k}^\mathsf{T} r_{k-1}\big), \end{aligned} \quad (24)$$

where (with identity matrix $I_{d_\beta} \in \mathbb{R}^{d_\beta \times d_\beta}$)

$$M_{k-1,k} = \mathrm{diag}\Big(I_{d_\beta}, K_{u_{k-1} u_k} K_{uu,k}^{-1}\Big), \quad (25)$$

$$\begin{aligned} R_{k-1} &= S_{k-1}^{-1} - \Lambda_{0,k-1}, \\ r_{k-1} &= S_{k-1}^{-1} m_{k-1} - \Lambda_{0,k-1} m_0. \end{aligned}$$

(R_{k-1}, r_{k-1}) are the natural parameters of the approximate likelihood of $\mathcal{D}_1, \ldots, \mathcal{D}_{k-1}$ (Hamelijnck et al., 2021), and are the only quantities stored from previous batches. Their blocks for β and u are

$$R_{k-1} = \begin{bmatrix} R_{\beta\beta,k-1} & R_{\beta u,k-1} \\ R_{u\beta,k-1} & R_{uu,k-1} \end{bmatrix}, \quad r_{k-1} = \begin{bmatrix} r_{\beta,k-1} \\ r_{u,k-1} \end{bmatrix}. \quad (26)$$

Under the GP prior, we also have $\mathbb{E}[\xi_{k-1} \mid \xi_k] = M_{k-1,k} \xi_k$. Detailed derivations are provided in Appendices B.1 and B.4.

3.2 Spatiotemporal inducing variables

The computation of the posterior in Eq. (24) involves inverting $K_{uu,k}$, where a naive approach would cost $\mathcal{O}(M_s^3 M_t^3)$ time. We address this by leveraging the separable spatiotemporal kernel (Eq. (2)) and constructing separate inducing variables for both spatial and temporal dimensions. This induces a Kronecker product structure for $K_{uu,k}$ which facilitates faster computations as discussed below.

In detail, for the spatial dimension, we fix M_s inducing locations $s_1^u, \ldots, s_{M_s}^u$. From the spatial kernel k_s, we define the covariance matrix K_s at these points as

$$[K_s]_{ij} = k_s(s_i^u, s_j^u), \quad 1 \le i, j \le M_s. \tag{27}$$

For each spatial inducing location s_j^u, we define M_t temporal inducing variables $u_{j,0,k}, \ldots, u_{j,M_t-1,k}$, where $u_{j,m,k}$ is the mth HiPPO projection of the GP $f(\tau, s_j^u)$ at step k. We define the initial time t_0 and the current time t_k for the kth update. Then the temporal inducing variables are defined as

$$u_{j,m,k} = \int_{t_0}^{t_k} f(\tau, s_j^u) g_{m,k}(\tau) \omega_k(\tau)\, d\tau, \tag{28}$$

where $\omega_k(\tau) = 1/(t_k - t_0)$ on $\tau \in [t_0, t_k]$, and

$$g_{m,k}(\tau) = \sqrt{2m+1}\, P_m\!\left(\frac{2(\tau - t_0)}{t_k - t_0} - 1\right), \tag{29}$$

with P_m the mth Legendre polynomial, so that $\int_{t_0}^{t_k} g_{m,k} g_{n,k}\, \omega_k\, d\tau = \delta_{mn}$. Stacking these coefficients into $U_k \in \mathbb{R}^{M_s \times M_t}$ with $[U_k]_{jm} = u_{j,m,k}$, we set $u_k = \mathrm{vec}(U_k)$, where vec stacks the columns of a matrix. With the separable kernel in Eq. (2), the prior covariance of u_k is then $K_{uu,k} = K_{t,k} \otimes K_s$, where $K_{t,k}$ is the covariance of the temporal HiPPO projections at step k, computed as in Eq. (10) with k_t. Similarly, $K_{u_{k-1}u_k} = K_{t,k-1,k} \otimes K_s$, where $K_{t,k-1,k}$ is the covariance between the projections at steps $k-1$ and k.

We now write A_k in Eq. (24) in Kronecker form. Assume that every location in a fixed spatial set $\mathcal{S}^{\mathrm{obs}}$ is observed at each observation time. The observation set for batch k is then $X_k = \mathcal{T}_k^{\mathrm{obs}} \times \mathcal{S}^{\mathrm{obs}}$, with $n_k = N_{t,k} N_s$ observations, where $N_{t,k} = |\mathcal{T}_k^{\mathrm{obs}}|$ and $N_s = |\mathcal{S}^{\mathrm{obs}}|$.

Let $K_{f_s u_s} \in \mathbb{R}^{N_s \times M_s}$ denote the spatial observation–inducing covariance, with $[K_{f_s u_s}]_{ij} = k_s(s_i^{\mathrm{obs}}, s_j^u)$, and let $K_{f_t u_t, k} \in \mathbb{R}^{N_{t,k} \times M_t}$ denote the corresponding temporal covariance, given by Eq. (9) for k_t. We order observations by time, then by spatial location. Under this observation layout, the separable kernel gives a Kronecker product form $K_{f_k u_k} = K_{f_t u_t, k} \otimes K_{f_s u_s}$ for the kernel matrix. Combined with the prior covariance Kronecker product factorization:

$$\begin{aligned} C_s &:= K_{f_s u_s} K_s^{-1}, \quad P_{t,k} := K_{f_t u_t, k} K_{t,k}^{-1}, \\ A_k &= K_{f_k u_k} K_{uu,k}^{-1} = P_{t,k} \otimes C_s, \\ A_k^{\mathsf{T}} A_k &= (P_{t,k}^{\mathsf{T}} P_{t,k}) \otimes (C_s^{\mathsf{T}} C_s). \quad \text{(required in } J_k^{\mathsf{T}} J_k) \end{aligned} \tag{30}$$

The spatial observation and inducing locations are fixed, so C_s does not depend on k and can be precomputed.

3.3 Analytical computation of HiPPO covariances via Bessel functions

Next, we describe how to compute the temporal covariance matrices $K_{t,k}$ and $K_{t,k-1,k}$ in Eqs. (30) and (25). We follow the RFF approximation of the temporal kernel in Eq. (12), whose HiPPO projections are computed by numerical ODE integration. Instead we show that the integrals can be evaluated analytically, which allows us to compute the covariance matrices efficiently.

By setting $t_0 = 0$ w.l.o.g., we denote $Z_k \in \mathbb{R}^{M_t \times D}, [Z_k]_{mj} := Z(t_k)_{mj}$ as a matrix collecting the recurrent states of the RFF features in Eq. (14). Applying a spherical Bessel function identity (Watson, 1944, § 3.32) to the integrals of sine and cosine functions against the Legendre basis gives the following analytical expression for $m = 0, \ldots, M_t - 1$:

$$\begin{bmatrix} [Z_k]_{mj} \\ [Z_k]_{m,j+D/2} \end{bmatrix} = \sqrt{\frac{2\sigma_f^2(2m+1)}{D}}\, j_m\!\left(\frac{\lambda_j(t_k - t_0)}{2}\right) \times \begin{bmatrix} \sin\!\left(\frac{\lambda_j(t_0+t_k)}{2} + \frac{m\pi}{2}\right) \\ \cos\!\left(\frac{\lambda_j(t_0+t_k)}{2} + \frac{m\pi}{2}\right) \end{bmatrix}. \tag{31}$$

where j_m is the spherical Bessel function of the first kind of order m.

We compute the spherical Bessel values in Eq. (31) with Miller's backward recurrence method (Bickley et al., 1952; Gautschi, 1967), which avoids the instability of forward recurrence when the order m is large relative to the magnitude of the argument $\lambda_j(t_k - t_0)/2$. This evaluates the analytical feature projection without integrating the HiPPO ODE; details are given in Appendix B.9, and a comparison with the ODE-based construction is given in Appendix C.8.

Stacking the RFF features $z(t)^{\mathsf{T}}$ of Eq. (13) at the observation times $t \in \mathcal{T}_k^{\mathrm{obs}}$ in temporal order into $Z_{X_k} \in \mathbb{R}^{N_{t,k} \times D}$, we obtain

$$K_{f_t u_t, k} \approx Z_{X_k} Z_k^{\mathsf{T}}, \quad K_{t,k} \approx Z_k Z_k^{\mathsf{T}}. \tag{32}$$

Again, $K_{t,k-1,k}$ is the temporal cross-covariance between the inducing variables at steps $k-1$ and k. Using the same frequencies for the old interval $[t_0, t_{k-1}]$ and the new interval $[t_0, t_k]$ gives

$$K_{t,k-1,k} \approx Z_{k-1} Z_k^{\mathsf{T}}. \tag{33}$$

$K_{t,k-1,k}$ is used in the online update of the next section.

3.4 Kronecker structure of the online update

We now substitute the covariance structure into the online update in Eq. (24). Using the GP block of

$M_{k-1,k}$ in Eq. (25), the natural parameters $R_{uu,k}$ and $r_{u,k}$ are updated as

$$\begin{aligned} R_{uu,k} &= \left(K_{u_{k-1}u_k} K_{uu,k}^{-1}\right)^{\mathsf{T}} R_{uu,k-1} \left(K_{u_{k-1}u_k} K_{uu,k}^{-1}\right) \\ &\quad + \sigma_y^{-2} A_k^{\mathsf{T}} A_k, \\ r_{u,k} &= \left(K_{u_{k-1}u_k} K_{uu,k}^{-1}\right)^{\mathsf{T}} r_{u,k-1} + \sigma_y^{-2} A_k^{\mathsf{T}} y_k. \end{aligned} \tag{34}$$

For both, the first term contains the contribution from previous batches, and the second is the contribution from the new batch.

Substituting the separable covariance matrices gives

$$\begin{aligned} K_{u_{k-1}u_k} K_{uu,k}^{-1} &= (K_{t,k-1,k} \otimes K_s)(K_{t,k} \otimes K_s)^{-1} \\ &= (K_{t,k-1,k} K_{t,k}^{-1}) \otimes I_{M_s}. \end{aligned} \tag{35}$$

In Appendix B.4 we further reduce the recurrence of $R_{uu,k-1}$ to $R_{uu,k-1} = B_{t,k-1} \otimes (C_s^{\mathsf{T}} C_s)$, where $B_{t,0} = 0$ and

$$B_{t,k} = K_{t,k}^{-1} K_{t,k-1,k}^{\mathsf{T}} B_{t,k-1} K_{t,k-1,k} K_{t,k}^{-1} + \sigma_y^{-2} P_{t,k}^{\mathsf{T}} P_{t,k} \tag{36}$$

We use this factorization for posterior computation in the next section. The complete derivation is given in Appendix B.4.

3.5 Efficient posterior computation and prediction

Substituting $R_{uu,k} = B_{t,k} \otimes (C_s^{\mathsf{T}} C_s)$ into Eq. (24) using the definitions in Eq. (26) gives the joint posterior precision

$$\begin{aligned} \Lambda_k &= \begin{bmatrix} S_{\beta,0}^{-1} + R_{\beta\beta,k} & R_{\beta u,k} \\ R_{u\beta,k} & D_{u,k} \end{bmatrix}, \\ D_{u,k} &= K_{t,k}^{-1} \otimes K_s^{-1} + B_{t,k} \otimes (C_s^{\mathsf{T}} C_s), \end{aligned} \tag{37}$$

Because $D_{u,k}$ is a sum of two Kronecker products, we solve linear systems involving this matrix using the whitening and eigendecomposition approach of Rakitsch et al. (2013, Section 3). The same algebra applies to the precision matrix here. The corresponding steps for our matrices are given in Appendix B.6.

Using these solves in the Schur complement formula (Cottle, 1974) gives the joint posterior mean $m_k = (m_{\beta,k}^{\mathsf{T}}, m_{u,k}^{\mathsf{T}})^{\mathsf{T}}$:

$$\begin{aligned} P_{\beta,k} &= S_{\beta,0}^{-1} + R_{\beta\beta,k} - R_{\beta u,k} D_{u,k}^{-1} R_{u\beta,k}, \\ m_{\beta,k} &= P_{\beta,k}^{-1} (S_{\beta,0}^{-1} m_{\beta,0} + r_{\beta,k} - R_{\beta u,k} D_{u,k}^{-1} r_{u,k}), \\ m_{u,k} &= D_{u,k}^{-1} (r_{u,k} - R_{u\beta,k} m_{\beta,k}). \end{aligned} \tag{38}$$

Each occurrence of $D_{u,k}^{-1}$ is evaluated by solving a linear system. The posterior covariance $S_k = \Lambda_k^{-1}$ follows from the same block inverse formula, given in Appendix B.7. These calculations are algebraically equivalent to solving the joint precision matrix directly.

For a new input x_*, let $k_{u*,k} = \mathrm{Cov}(u_k, f(x_*))$. The predictive mean and variance of $y_* = y(x_*)$ are

$$\begin{aligned} a_{*,k} &= K_{uu,k}^{-1} k_{u*,k}, \quad h_{*,k} = \begin{bmatrix} \phi(x_*) \\ a_{*,k} \end{bmatrix}, \\ \nu_{*,k} &= k(x_*, x_*) - a_{*,k}^{\mathsf{T}} K_{uu,k} a_{*,k}, \\ \mathbb{E}_{q_k}[y_*] &= \phi(x_*)^{\mathsf{T}} m_{\beta,k} + a_{*,k}^{\mathsf{T}} m_{u,k}, \\ \mathrm{Var}_{q_k}(y_*) &= \sigma_y^2 + \nu_{*,k} + h_{*,k}^{\mathsf{T}} S_k h_{*,k}. \end{aligned} \tag{39}$$

The term $h_{*,k}^{\mathsf{T}} S_k h_{*,k}$ accounts for uncertainty in the trend and GP residual, including their cross-covariance. It can be evaluated with the same linear solves without storing the full posterior covariance.

With fixed trend dimension d_β, posterior computation costs $\mathcal{O}(M_s^3 + M_t^3)$ time and $\mathcal{O}(M_s^2 + M_t^2 + M_s M_t)$ memory. For $M_s = M_t = M$, these costs become $\mathcal{O}(M^3)$ time and $\mathcal{O}(M^2)$ memory, compared with $\mathcal{O}(M^6)$ time and $\mathcal{O}(M^4)$ memory for dense computation.

The dependence on d_β is given in Appendix B.8.

4 EXPERIMENTS

We evaluate KronHiPPO-STGP on three online spatiotemporal prediction tasks, ablate its components, and compare Kronecker-structured and dense posterior computation. Details are given in Appendix C.

Experimental setup. At each online step, predictions are made using only observations available before the prediction time. Labels revealed after prediction are used only for subsequent online updates.

We compare KronHiPPO-STGP with streaming sparse GPs (Bui et al., 2017; Maddox et al., 2021), OHSVGP (Chen et al., 2025), structured spatiotemporal GP baselines where applicable, and IGNNK for spatiotemporal kriging on PEMS-BAY (Wu et al., 2021). We report RMSE, CRPS, Gaussian NLPD, and ECE. Lower values indicate better performance. Results are averaged over spatial splits with sample standard deviations. To assess the effect of using HiPPO projection for the temporal inducing variables, we compare with Kron-STGP, which uses the same trend–residual model, but replaces the evolving HiPPO basis with a fixed full-horizon temporal basis. The difference between the two methods isolates the effect of using HiPPO interdomain representation for temporal inducing variables.

Online prediction. Figure 1 compares the methods across the three datasets. The largest RMSE reduction occurs on ERA5-Land: KronHiPPO-STGP reduces RMSE by 66% relative to Kron-STGP.

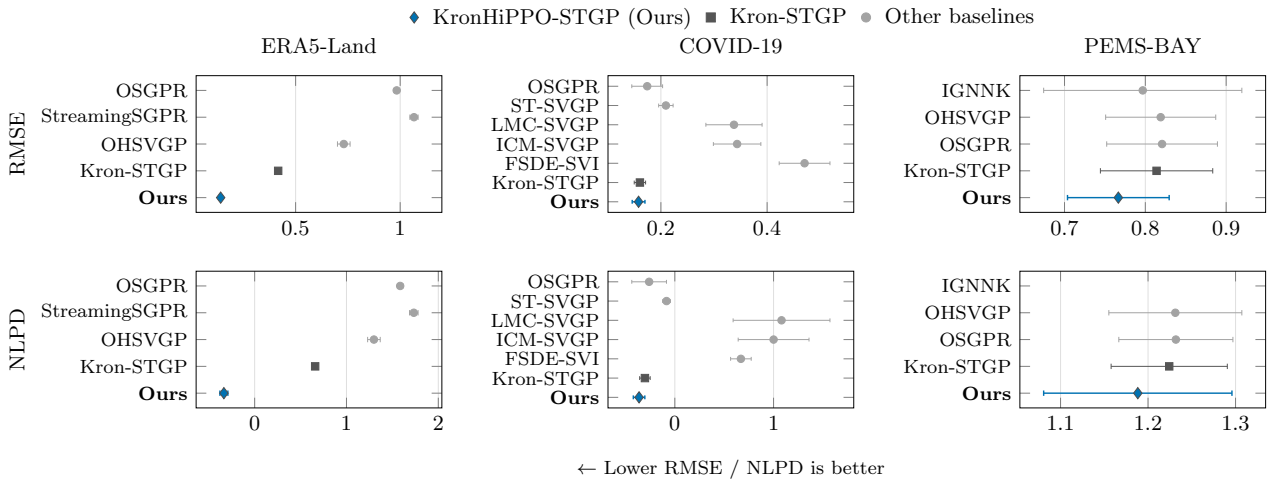


Figure 1: Online prediction across three spatiotemporal datasets. Markers and horizontal bars show means and ±1 sample standard deviation over spatial splits. Metrics unavailable for IGNNK on PEMS-BAY are omitted. Blue diamonds denote KronHiPPO-STGP and dark squares denote Kron-STGP.

On COVID-19, the RMSE difference relative to Kron-STGP is small (0.1605 → 0.1580), while CRPS and NLPD improve more clearly (0.0922 → 0.0885 and −0.3029 → −0.3621, respectively). On PEMS-BAY, KronHiPPO-STGP reduces RMSE by 5.8% relative to Kron-STGP and by 3.8% relative to IGNNK. Figure 2 shows representative PEMS-BAY trajectories.

Ablation studies. Table 1 reports ablations of the off-diagonal blocks $R_{\beta u}$ and $R_{u\beta}$ and the evolving HiPPO basis.

Setting $R_{\beta u}$ and $R_{u\beta}$ to zero removes the coupling between the trend coefficients β and the inducing variables u, so updates of β no longer affect the inference of u, and vice versa. This causes the largest degradation in RMSE, CRPS, and NLPD on both datasets: the RMSE increases about tenfold on ERA5-Land (0.1251 → 1.2898) and by 25% on PEMS-BAY (0.7666 → 0.9555). This is a failure similar to the mean-field approximation discussed in Section 3, and shows that the coupling between the trend and residual is essential for accurate online inference. Replacing the evolving HiPPO basis with a fixed full-horizon basis causes a smaller but consistent degradation, increasing the RMSE by 56% on ERA5-Land (0.1251 → 0.1950) and by 5.8% on PEMS-BAY (0.7666 → 0.8112). Full results are in Appendix C.5.

Table 1: Ablations of the off-diagonal blocks and the temporal basis. "w/o $R_{\beta u}$" sets $R_{\beta u}$ and $R_{u\beta}$ to zero; "w/ full-horizon basis" replaces the evolving HiPPO basis with a fixed temporal basis. Lower is better.

Variant	RMSE ↓	CRPS ↓	NLPD ↓	ECE ↓
ERA5-Land				
Full model	**0.1251**	**0.0665**	**−0.6837**	**0.0461**
w/o $R_{\beta u}$	1.2898	0.9951	52.8139	0.6905
w/ full-horizon basis	0.1950	0.1420	0.2580	0.1807
PEMS-BAY				
Full model	**0.7666**	**0.3774**	**1.1883**	0.1317
w/o $R_{\beta u}$	0.9555	0.4856	1.5333	**0.0474**
w/ full-horizon basis	0.8112	0.4092	1.2194	0.1760

Temporal inducing variables across online updates. In Eq. (24), we project the information from the previous batches (R_{k-1}, r_{k-1}) from u_{k-1} to u_k using $M_{k-1,k}$, corresponding to the scale change in temporal basis. On ERA5-Land, we compare this update with a variant that replaces $M_{k-1,k}$ with the identity matrix. As a reference, we use the variational posterior recomputed from all observations so far. At $M_t = 128$, the predictive KL relative to the reference is 1.26×10^{-4} with the projection $M_{k-1,k}$, and 2.07 without projection; RMSE is 0.0795 and 0.2651, respectively (Table 6). Without projection, KL remains order one for all $M_t \geq 32$ (Figure 3(a)), and increasing the number of temporal inducing variables does not compensate for this error. Detailed results are provided in Appendix C.5 (Figure 5).

Kronecker-structured posterior computation. We compare the Kronecker-structured posterior computation in Section 3.5 with dense factorization on the same positive-definite Gaussian precision systems. At $M_s = M_t = 64$, the structured solver is 1036.7× faster than dense factorization while retaining relative solution error below 5×10^{-12} (Table 2). Detailed timing

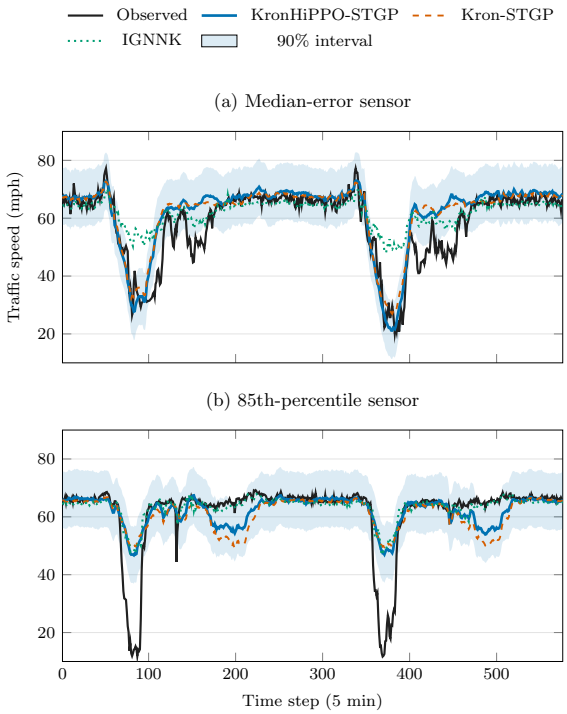


Figure 2: PEMS-BAY predictions for two representative held-out sensors. Observed and predicted trajectories are shown for KronHiPPO-STGP, IGNNK, and Kron-STGP; shading denotes the 90% predictive interval of KronHiPPO-STGP.

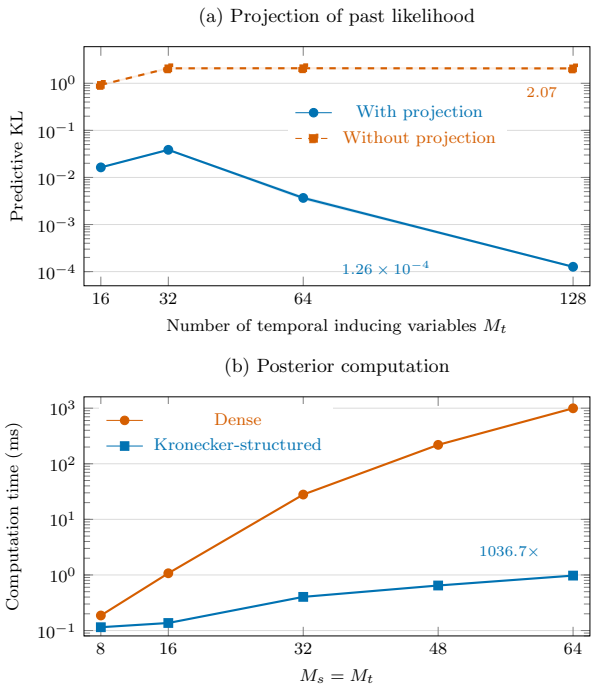


Figure 3: Diagnostics for online updates and posterior computation. (a) Predictive KL with and without the projection $M_{k-1,k}$ across M_t (see also Figure 5). (b) Dense and Kronecker-structured posterior computation time.

results are provided in Appendix C.6.

5 DISCUSSION

Related work. Hamelijnck et al. (2021) proposed ST-VGP and its spatially sparse approximation ST-SVGP which use a variational inference with Markov temporal kernel via Kalman filtering and smoothing. Their batch learning processes the full time series at each optimization iteration, with sequential computational cost linear in the number of time steps. Our method updates the model online using fixed-size memory without accessing previous observations. For fixed batch and model sizes, the computational cost per update is independent of the history length. We observe significantly faster training, as described in Appendix C.6. Furthermore, our method jointly estimates linear trend coefficients and supports stationary temporal kernels whose spectral densities can be sampled for the RFF approximation.

Table 2: Dense versus Kronecker-structured posterior solve. Relative error is measured against the dense solution.

$M_s = M_t$	Dense	Struct.	Speed-up ↑	Rel. err. ↓
16	1.072	0.137	7.9×	9.2×10^{-14}
32	27.919	0.403	69.6×	9.1×10^{-13}
64	994.822	0.974	1036.7×	4.7×10^{-12}

Conclusion. We proposed KronHiPPO-STGP, an online Bayesian spatiotemporal model that extends the online HiPPO-SVGP with Kronecker-structured GP inference and a Bayesian linear term that captures global trends with fixed memory. By utilizing closed-form solution, we avoid numerical integration of the HiPPO ODEs, and by exploiting the Kronecker structure, we enable efficient posterior computation. In the experiments, our method produced accurate predictions while requiring less training time than the spatiotemporal variational GP baselines.

Limitation and future work. Our current Kronecker-structured computation assumes a separable kernel, a Gaussian likelihood, the same spatial locations at every observation time, and hyperparameters fixed after the initial task. Future work includes supporting time-varying observation locations and updating the hyperparameters online.

AI use statement

We used OpenAI ChatGPT and Codex as research-support and editing tools. Their use included suggesting literature-search queries and organizing candidate sources; assisting with revisions to experimental code and additions to training scripts; organizing experiment records and checking the consistency of tables and reported results; improving LaTeX layout, cross-references, and figure and table formatting; and suggesting language and structural revisions to selected sections. Generative-AI outputs were treated as suggestions rather than scholarly evidence. The authors checked cited claims against the original publications, reviewed and tested code changes, generated the reported numerical results by running the documented experiments, and verified mathematical arguments and reported values against project records. All material was selected, revised, and approved by the authors, who remain responsible for its accuracy and originality.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes] Section 3 and Appendix B.
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes] Section 3.5 and Appendix B.8.
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [No] If reviewers request code, we will provide an anonymized archive through OpenReview. We will release the code upon acceptance.

2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. [Yes] Section 3 and

Appendix B.8. The results assume a Gaussian likelihood, a separable spatiotemporal kernel, fixed spatial inducing locations, and the same spatial locations observed at each observation time.

(b) Complete proofs of all theoretical results. [Yes] Appendix B.

(c) Clear explanations of any assumptions. [Yes] Sections 2.1 and 3.2.

3. For all figures and tables that present empirical results, check if you include:

(a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [No] The datasets are public and cited, and the preprocessing is described in Appendix C.3. We will release the code upon acceptance.

(b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes] Appendices C.2 and C.3.

(c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes] Appendix C.1; error bars in the main results are sample standard deviations over spatial splits, and runtime error bars are over repeated runs.

(d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes] Appendix C.6. The dense versus structured solver comparison and some other experiments used an NVIDIA GeForce RTX 5070 Laptop GPU; all other timing measurements used a single NVIDIA GeForce RTX 4090.

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:

(a) Citations of the creator If your work uses existing assets. [Yes] Section 4 and Appendix C.3.

(b) The license information of the assets, if applicable. [Yes] Appendix C.3.

(c) New assets either in the supplemental material or as a URL, if applicable. [Not Applicable]

(d) Information about consent from data providers/curators. [Not Applicable] We use only publicly released datasets.

(e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable] The COVID-19 data are aggregated hospital admissions and contain no individual-level records.

5. If you used crowdsourcing or conducted research with human subjects, check if you include:

(a) The full text of instructions given to participants and screenshots. [Not Applicable]

(b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]

(c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

KronHiPPO-STGP: Online Bayesian Spatiotemporal Modeling with Evolving Memory: Supplementary Materials

A NOTATION

Table 3 summarizes the notation used in the main text. For random vectors a and b, $K_{ab} = \mathrm{Cov}(a, b)$ under the GP prior, and we write $K_{uu,k} := K_{u_k u_k}$ for brevity. Kronecker factors carry the subscript s (space) or t (time), and the subscript k denotes the online step.

Table 3: Notation.

Symbol	Meaning
Model	
$x = (t, s)$, y	Input (time, location) and observation
$\phi(t, s) \in \mathbb{R}^{d_\beta}$, β	Trend basis and trend coefficients
$m_{\beta,0}$, $S_{\beta,0}$	Prior mean and covariance of β
f, $k = k_t k_s$	GP residual and its separable kernel
σ_y^2, σ_f^2	Observation noise variance and kernel variance
Background (Sections 2.2 and 2.3)	
$\widetilde{x}_m$, u, M	Inducing points, inducing variables, and their number
m_u, S_u, $\mathcal{F}$	Variational mean and covariance of u, and the VFE objective
ψ_m	Basis function of an interdomain inducing variable
$g_m^{(t)}$, $\omega^{(t)}$, $\psi_m^{(t)} = g_m^{(t)}\omega^{(t)}$	HiPPO basis and measure on $[0, t]$
$K_{uu}^{(t)}$, $K_{uf}^{(t)}$	HiPPO covariances at time t for a one-dimensional GP
$p(\lambda)$, h	Spectral density of a stationary kernel and time difference
Online setting	
$\mathcal{D}_k$, n_k, K	kth batch, its size, and the number of batches
$x_{ki} = (t_{ki}, s_{ki})$, y_{ki}	ith input and observation in $\mathcal{D}_k$
y_k, f_k	Observations in $\mathcal{D}_k$ and the GP values $f_{ki} = f(x_{ki})$ at their inputs
t_0, t_k	Initial time and current time at step k
HiPPO inducing variables	
M_s, M_t	Numbers of spatial inducing points and temporal inducing variables
s_j^u	jth spatial inducing location
P_m, $g_{m,k}$, ω_k	Legendre polynomial, scaled Legendre basis on $[t_0, t_k]$, and uniform measure
$u_{j,m,k}$	mth HiPPO projection of $f(\cdot, s_j^u)$ at step k
$U_k \in \mathbb{R}^{M_s \times M_t}$, $u_k = \mathrm{vec}(U_k)$	Inducing variables at step k
$A_{\mathrm{H}}(t)$, $b_{\mathrm{H}}(t)$	HiPPO ODE coefficients
Random Fourier features and covariances	
D, λ_j, $z(\tau)$	Number of random features, sampled frequencies, and feature vector
$Z(t)$, $Z_k = Z(t_k) \in \mathbb{R}^{M_t \times D}$	HiPPO projections of the features
Z_{X_k}	Features $z(t)^\mathsf{T}$ stacked over the observation times in $\mathcal{D}_k$
j_m	Spherical Bessel function of the first kind of order m
K_s, $K_{t,k}$	Spatial and temporal prior covariance factors; $[K_{t,k}]_{mn} = \iint \psi_{m,k}(\tau) k_t(\tau, \tau') \psi_{n,k}(\tau')\, d\tau\, d\tau' \approx [Z_k Z_k^\mathsf{T}]_{mn}$ with $\psi_{m,k} = g_{m,k}\omega_k$
$K_{uu,k} = K_{u_k u_k}$	Prior covariance of u_k, $K_{uu,k} = K_{t,k} \otimes K_s$
$K_{u_{k-1} u_k}$, $K_{t,k-1,k}$	Cross-covariance between steps $k-1$ and k, $K_{u_{k-1} u_k} = K_{t,k-1,k} \otimes K_s$ with $[K_{t,k-1,k}]_{mn} = \iint \psi_{m,k-1}(\tau) k_t(\tau, \tau') \psi_{n,k}(\tau')\, d\tau\, d\tau' \approx [Z_{k-1} Z_k^\mathsf{T}]_{mn}$
$K_{f_k u_k}$	Covariance between f_k and u_k, $K_{f_k u_k} = K_{f_t u_t, k} \otimes K_{f_s u_s}$

Symbol	Meaning
Online update	
$\xi_k = (\beta, u_k)$	Joint state
$p_k(\xi_k) = \mathcal{N}(m_0, \Lambda_{0,k}^{-1})$	Prior on the joint state
$q_k(\xi_k) = \mathcal{N}(m_k, S_k)$	Approximate posterior at step k, with $m_k = (m_{\beta,k}, m_{u,k})$
$\widetilde{q}_k(\xi_{k-1})$	q_k mapped to ξ_{k-1} through $p_k(u_{k-1} \mid u_k)$
$\mathcal{L}_k^{\text{joint}}$	Online ELBO for the joint state
Φ_k, A_k, $J_k = [\Phi_k\ A_k]$	Trend features, $A_k = K_{f_k u_k} K_{uu,k}^{-1}$, and joint design matrix
$M_{k-1,k}$	Map from ξ_k to $\mathbb{E}[\xi_{k-1} \mid \xi_k]$ under the GP prior
R_k, r_k	Natural parameters of the approximate likelihood of $\mathcal{D}_1, \dots, \mathcal{D}_k$; blocks $R_{\beta\beta}$, $R_{\beta u}$, $R_{u\beta}$, R_{uu} and r_β, r_u
I_{d_β}, I_{M_s}, I_{n_k}	Identity matrices of the indicated sizes
Kronecker computation	
$\mathcal{S}^{\text{obs}}$, $\mathcal{T}_k^{\text{obs}}$, X_k	Observed locations, observation times in $\mathcal{D}_k$, and $X_k = \mathcal{T}_k^{\text{obs}} \times \mathcal{S}^{\text{obs}}$; $N_s = \lvert\mathcal{S}^{\text{obs}}\rvert$, $N_{t,k} = \lvert\mathcal{T}_k^{\text{obs}}\rvert$
s_i^{obs}	ith observed location
$K_{f_s u_s}$, $K_{f_t u_t,k}$	Spatial and temporal observation–inducing covariances; $[K_{f_t u_t,k}]_{im} = \int k_t(t_i, \tau)\psi_{m,k}(\tau)\, d\tau \approx [Z_{X_k} Z_k^\mathsf{T}]_{im}$ for $t_i \in \mathcal{T}_k^{\text{obs}}$
C_s, $P_{t,k}$	$K_{f_s u_s} K_s^{-1}$ and $K_{f_t u_t,k} K_{t,k}^{-1}$, so that $A_k = P_{t,k} \otimes C_s$
$B_{t,k}$	Temporal factor of $R_{uu,k} = B_{t,k} \otimes (C_s^\mathsf{T} C_s)$, with $B_{t,0} = 0$
Λ_k, $D_{u,k}$	Joint posterior precision and its residual block
$P_{\beta,k}$	Schur complement for the trend block
Prediction	
x_*, $k_{u*,k}$	Test input and $\text{Cov}(u_k, f(x_*))$
$a_{*,k}$, $h_{*,k}$, $\nu_{*,k}$	$K_{uu,k}^{-1} k_{u*,k}$, $(\phi(x_*), a_{*,k})$, and the conditional variance of $f(x_*)$ given u_k

B DERIVATION OF THE ONLINE UPDATE

This appendix derives the online update of KronHiPPO-STGP. We derive (i) the Gaussian likelihood of each batch in natural-parameter form, (ii) how the likelihood of previous batches is expressed in terms of the current inducing variables when the HiPPO basis changes, and (iii) the posterior computation that exploits the Kronecker structure.

B.1 Gaussian batch likelihood in natural-parameter form

Consider the k-th batch with inputs $x_{k1}, \dots, x_{kn_k}$ and outputs $y_k = (y_{k1}, \dots, y_{kn_k})^\mathsf{T}$. Let $f_k = (f(x_{k1}), \dots, f(x_{kn_k}))^\mathsf{T}$ and $[\Phi_k]_{i,:} = \phi(x_{ki})^\mathsf{T}$. The observation model and the conditional GP distribution given the inducing variables are

$$\begin{aligned} p(y_k|\beta, f_k) &= \mathcal{N}(\Phi_k\beta + f_k, \sigma_y^2 I_{n_k}), \\ p(f_k|u_k) &= \mathcal{N}(A_k u_k, C_{f|u,k}), \end{aligned} \tag{40}$$

where $A_k = K_{f_k u_k} K_{uu,k}^{-1}$ and $C_{f|u,k} = K_{f_k f_k} - A_k K_{uu,k} A_k^\mathsf{T}$. Let $\xi_k = (\beta^\mathsf{T}, u_k^\mathsf{T})^\mathsf{T}$ and $J_k = [\Phi_k\ A_k]$. Writing $\epsilon_k = f_k - A_k u_k$, which has mean zero and covariance $C_{f|u,k}$ given u_k, the expected log-likelihood is

$$\begin{aligned} \mathbb{E}_{p(f_k|u_k)}[\log p(y_k|\beta, f_k)] &= -\frac{1}{2\sigma_y^2}\|y_k - J_k\xi_k\|^2 - \frac{1}{2\sigma_y^2}\operatorname{tr}(C_{f|u,k}) + c \\ &= -\frac{1}{2}\xi_k^\mathsf{T} R_k^{\text{new}} \xi_k + (r_k^{\text{new}})^\mathsf{T}\xi_k + c_k, \end{aligned} \tag{41}$$

with

$$R_k^{\text{new}} = \sigma_y^{-2}\begin{bmatrix} \Phi_k^\mathsf{T}\Phi_k & \Phi_k^\mathsf{T} A_k \\ A_k^\mathsf{T}\Phi_k & A_k^\mathsf{T} A_k \end{bmatrix}, \qquad r_k^{\text{new}} = \sigma_y^{-2}\begin{bmatrix} \Phi_k^\mathsf{T} y_k \\ A_k^\mathsf{T} y_k \end{bmatrix}. \tag{42}$$

We denote the blocks of R by $R_{\beta\beta}$, $R_{\beta u}$, $R_{u\beta}$, R_{uu} and those of r by r_β, r_u. For independent batches expressed in the same inducing variables, the natural parameters add:

$$R_{1:k} = \sum_{j=1}^{k} R_j^{\text{new}}, \qquad r_{1:k} = \sum_{j=1}^{k} r_j^{\text{new}}. \tag{43}$$

The online update therefore stores only (R_k, r_k) instead of the past observations; it reduces to Eq. (43) when the inducing variables do not change.

B.2 Necessity of the off-diagonal blocks

After conditioning on past observations, the trend β and the residual u are correlated. When the trend coefficients change, the residual natural parameter conditioned on them changes accordingly. We show that this correction requires the off-diagonal blocks $R_{\beta u}$ and $R_{u\beta}$ of R.

Let (R_{k-1}, r_{k-1}) denote the natural parameters of the approximate likelihood of the batches before update k. For a fixed trend value β, the residual natural parameter is $\zeta_{u,k-1}(\beta) = r_{u,k-1} - R_{u\beta,k-1}\beta$. If a later update changes β to $\beta + \Delta\beta$, then

$$\Delta\zeta_{u,k-1} = \zeta_{u,k-1}(\beta + \Delta\beta) - \zeta_{u,k-1}(\beta) = -R_{u\beta,k-1}\Delta\beta. \tag{44}$$

Proposition 1 (Necessity of the off-diagonal blocks). Suppose that past observations are summarized by natural parameters and not stored. For the residual natural parameter to be updated exactly for every change $\Delta\beta$ of the trend posterior, the stored statistics must include $R_{u\beta,k-1}$ (or any statistic that determines the map $\Delta\beta \mapsto -R_{u\beta,k-1}\Delta\beta$). In particular, $(R_{uu,k-1}, r_{u,k-1})$ alone is insufficient.

Proof. Consider two sets of past observations with the same $(R_{uu,k-1}, r_{u,k-1})$ but $R_{u\beta}^{(1)} \neq R_{u\beta}^{(2)}$. Then there exists $\Delta\beta$ with $R_{u\beta}^{(1)}\Delta\beta \neq R_{u\beta}^{(2)}\Delta\beta$, so by Eq. (44) the required corrections differ. An update that uses only $(R_{uu,k-1}, r_{u,k-1})$ gives the same result in both cases and therefore cannot be exact for both. □

We therefore store the natural parameters of the approximate likelihood over (β, u), including $R_{u\beta}$.

B.3 Inducing variables under a changing HiPPO basis

The HiPPO inducing variables are defined on a time interval that grows with each update, so the inducing variables at the previous update and those at the current update are different random variables. Reusing the previous natural parameters directly would treat them as the same. Instead, we use their joint Gaussian prior.

For $k \geq 2$, the inducing variables u_{k-1} and u_k at the previous and current updates have joint GP prior

$$\begin{bmatrix} u_{k-1} \\ u_k \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} K_{uu,k-1} & K_{u_{k-1}u_k} \\ K_{u_k u_{k-1}} & K_{uu,k} \end{bmatrix}\right). \tag{45}$$

Gaussian conditioning gives

$$u_{k-1} = K_{u_{k-1}u_k} K_{uu,k}^{-1} u_k + \delta, \qquad \delta \sim \mathcal{N}(0, \Sigma_{k-1|k}), \tag{46}$$

where $\Sigma_{k-1|k} = K_{uu,k-1} - K_{u_{k-1}u_k} K_{uu,k}^{-1} K_{u_k u_{k-1}}$. Here $K_{u_{k-1}u_k} K_{uu,k}^{-1} u_k$ is the part of u_{k-1} predictable from u_k, and δ is the part that u_k cannot explain.

B.4 Approximate likelihood of previous batches in the current inducing variables

Using the approximate likelihood of previous batches $\ell_{k-1}(\xi_{k-1}) \propto q_{k-1}(\xi_{k-1})/p_{k-1}(\xi_{k-1})$ from Eq. (23), we write

$$\log \ell_{k-1}(\xi_{k-1}) = -\frac{1}{2}\xi_{k-1}^{\mathsf{T}} R_{k-1}\xi_{k-1} + r_{k-1}^{\mathsf{T}}\xi_{k-1} + c, \qquad \xi_{k-1} = \begin{bmatrix} \beta \\ u_{k-1} \end{bmatrix}, \tag{47}$$

and let $\xi_k = (\beta^{\mathsf{T}}, u_k^{\mathsf{T}})^{\mathsf{T}}$. By Eq. (46),

$$\xi_{k-1} = M_{k-1,k}\xi_k + e, \qquad M_{k-1,k} = \operatorname{diag}\left(I_{d_\beta}, K_{u_{k-1}u_k} K_{uu,k}^{-1}\right), \qquad e = \begin{bmatrix} 0 \\ \delta \end{bmatrix}. \tag{48}$$

Substituting this into $\log \ell_{k-1}$ gives

$$\begin{aligned}\log \ell_{k-1}(\xi_{k-1}) = &-\frac{1}{2}\xi_k^\mathsf{T} M_{k-1,k}^\mathsf{T} R_{k-1} M_{k-1,k}\xi_k - \xi_k^\mathsf{T} M_{k-1,k}^\mathsf{T} R_{k-1} e - \frac{1}{2} e^\mathsf{T} R_{k-1} e \\ &+ r_{k-1}^\mathsf{T} M_{k-1,k}\xi_k + r_{k-1}^\mathsf{T} e + c.\end{aligned} \tag{49}$$

Since $\mathbb{E}[e|\xi_k] = 0$, the terms linear in e vanish in conditional expectation, and $\mathbb{E}[e^\mathsf{T} R_{k-1} e|\xi_k] = \operatorname{tr}(R_{uu,k-1}\Sigma_{k-1|k})$ does not depend on ξ_k. Hence

$$\mathbb{E}[\log \ell_{k-1}(\xi_{k-1})|\xi_k] = -\frac{1}{2}\xi_k^\mathsf{T} R_{k-1\to k}\xi_k + r_{k-1\to k}^\mathsf{T}\xi_k + c', \tag{50}$$

with

$$\begin{aligned}R_{k-1\to k} &= M_{k-1,k}^\mathsf{T} R_{k-1} M_{k-1,k} \\ &= \begin{bmatrix} R_{\beta\beta,k-1} & R_{\beta u,k-1} K_{u_{k-1}u_k} K_{uu,k}^{-1} \\ \left(K_{u_{k-1}u_k} K_{uu,k}^{-1}\right)^\mathsf{T} R_{u\beta,k-1} & \left(K_{u_{k-1}u_k} K_{uu,k}^{-1}\right)^\mathsf{T} R_{uu,k-1} K_{u_{k-1}u_k} K_{uu,k}^{-1} \end{bmatrix}, \\ r_{k-1\to k} &= M_{k-1,k}^\mathsf{T} r_{k-1} = \begin{bmatrix} r_{\beta,k-1} \\ \left(K_{u_{k-1}u_k} K_{uu,k}^{-1}\right)^\mathsf{T} r_{u,k-1} \end{bmatrix}.\end{aligned} \tag{51}$$

Here $R_{k-1\to k}$ and $r_{k-1\to k}$ contain only the contribution from previous batches, expressed in the current variables. Adding the new-batch terms in Eq. (42) gives $R_k = R_{k-1\to k} + R_k^{\text{new}}$ and $r_k = r_{k-1\to k} + r_k^{\text{new}}$. The blocks of R_{k-1} and r_{k-1} follow Eq. (26). Since the conditional expectation under the GP prior is the orthogonal projection onto the L^2 functions of ξ_k, Eq. (50) minimizes the mean-square error to $\log \ell_{k-1}(\xi_{k-1})$ among all functions of ξ_k.

Kronecker structure of the GP precision block. Assume $R_{uu,k-1} = B_{t,k-1} \otimes (C_s^\mathsf{T} C_s)$. Substituting $A_k = P_{t,k} \otimes C_s$ and $K_{u_{k-1}u_k} K_{uu,k}^{-1} = (K_{t,k-1,k} K_{t,k}^{-1}) \otimes I_{M_s}$ from Eqs. (30) and (35) into the GP block of the update gives

$$\begin{aligned}R_{uu,k} &= \left[(K_{t,k-1,k} K_{t,k}^{-1}) \otimes I_{M_s}\right]^\mathsf{T} \left[B_{t,k-1} \otimes (C_s^\mathsf{T} C_s)\right] \left[(K_{t,k-1,k} K_{t,k}^{-1}) \otimes I_{M_s}\right] \\ &\quad + \sigma_y^{-2} (P_{t,k}^\mathsf{T} P_{t,k}) \otimes (C_s^\mathsf{T} C_s) \\ &= \left(K_{t,k}^{-1} K_{t,k-1,k}^\mathsf{T} B_{t,k-1} K_{t,k-1,k} K_{t,k}^{-1} + \sigma_y^{-2} P_{t,k}^\mathsf{T} P_{t,k}\right) \otimes (C_s^\mathsf{T} C_s) \\ &= B_{t,k} \otimes (C_s^\mathsf{T} C_s),\end{aligned} \tag{52}$$

where $B_{t,k}$ is defined in Eq. (36). Both contributions have the same spatial factor $C_s^\mathsf{T} C_s$, which is fixed across updates. Before the first batch, there is no likelihood contribution, so $R_{uu,0} = 0$ and $B_{t,0} = 0$. The first update therefore gives $B_{t,1} = \sigma_y^{-2} P_{t,1}^\mathsf{T} P_{t,1}$ without a historical term. The factorization then holds at every subsequent update by induction.

B.5 Error of the conditional expectation

The conditional expectation in Eq. (50) is optimal but not exact in general, because u_{k-1} contains a component δ that u_k cannot explain. We compute the resulting mean-square error.

Let $g(\xi_{k-1}) = -\frac{1}{2}\xi_{k-1}^\mathsf{T} R_{k-1}\xi_{k-1} + r_{k-1}^\mathsf{T}\xi_{k-1}$ and $\epsilon_g = g(\xi_{k-1}) - \mathbb{E}[g(\xi_{k-1})|\xi_k]$. Using Eq. (48),

$$g(\xi_{k-1}) = g(M_{k-1,k}\xi_k) + a^\mathsf{T}\delta - \frac{1}{2}\delta^\mathsf{T} R_{uu,k-1}\delta, \qquad a = r_{u,k-1} - R_{u\beta,k-1}\beta - R_{uu,k-1} K_{u_{k-1}u_k} K_{uu,k}^{-1} u_k, \tag{53}$$

so that

$$\epsilon_g = a^\mathsf{T}\delta - \frac{1}{2}\left(\delta^\mathsf{T} R_{uu,k-1}\delta - \operatorname{tr}(R_{uu,k-1}\Sigma_{k-1|k})\right). \tag{54}$$

Since $\delta \mid \xi_k \sim \mathcal{N}(0, \Sigma_{k-1|k})$, $\operatorname{Var}(a^\mathsf{T}\delta \mid \xi_k) = a^\mathsf{T}\Sigma_{k-1|k} a$ and $\operatorname{Var}(\delta^\mathsf{T} R_{uu,k-1}\delta \mid \xi_k) = 2\operatorname{tr}(R_{uu,k-1}\Sigma_{k-1|k} R_{uu,k-1}\Sigma_{k-1|k})$, and the two terms are conditionally uncorrelated because the third moments of a zero-mean Gaussian vanish. Therefore

$$\mathbb{E}[\epsilon_g^2|\xi_k] = a^\mathsf{T}\Sigma_{k-1|k} a + \frac{1}{2}\operatorname{tr}\left((R_{uu,k-1}\Sigma_{k-1|k})^2\right). \tag{55}$$

Both terms vanish when $\Sigma_{k-1|k} = 0$, that is, when u_{k-1} is a deterministic function of u_k.

B.6 Posterior computation

We now compute the posterior without forming the full $M_tM_s \times M_tM_s$ covariance matrix. By Section 3.4, $R_{uu,k} = B_{t,k} \otimes (C_s^\mathsf{T} C_s)$. Adding the prior precision and the likelihood natural parameters, the joint posterior precision is

$$\Lambda_k = \begin{bmatrix} S_{\beta,0}^{-1} + R_{\beta\beta,k} & R_{\beta u,k} \\ R_{u\beta,k} & D_{u,k} \end{bmatrix}, \tag{56}$$

where

$$D_{u,k} = K_{t,k}^{-1} \otimes K_s^{-1} + B_{t,k} \otimes (C_s^\mathsf{T} C_s). \tag{57}$$

The residual block is a sum of two Kronecker products. A direct inversion would not exploit this structure and would cost $O(M_t^3 M_s^3)$. Instead, we solve linear systems involving $D_{u,k}$.

Sylvester form. For $V, Q \in \mathbb{R}^{M_s \times M_t}$, the identity $(A \otimes B)\,\mathrm{vec}(V) = \mathrm{vec}(BVA^\mathsf{T})$ and the symmetry of $K_{t,k}$ and $B_{t,k}$ give

$$D_{u,k}\,\mathrm{vec}(V) = \mathrm{vec}(Q) \quad \Longleftrightarrow \quad K_s^{-1} V K_{t,k}^{-1} + C_s^\mathsf{T} C_s V B_{t,k} = Q. \tag{58}$$

Whitening and eigendecomposition. Let $K_s^{1/2}$ and $K_{t,k}^{1/2}$ denote symmetric positive-definite square roots and set $V = K_s^{1/2} X K_{t,k}^{1/2}$. Substituting into Eq. (58) and multiplying by $K_s^{1/2}$ on the left and $K_{t,k}^{1/2}$ on the right gives

$$X + \widetilde{G}_s X \widetilde{B}_t = \widetilde{Q}, \tag{59}$$

where $\widetilde{G}_s = K_s^{1/2} C_s^\mathsf{T} C_s K_s^{1/2}$, $\widetilde{B}_t = K_{t,k}^{1/2} B_{t,k} K_{t,k}^{1/2}$, and $\widetilde{Q} = K_s^{1/2} Q K_{t,k}^{1/2}$. Both $\widetilde{G}_s$ and $\widetilde{B}_t$ are symmetric positive semidefinite. With the eigendecompositions $\widetilde{G}_s = U_s \Gamma U_s^\mathsf{T}$ and $\widetilde{B}_t = U_t H U_t^\mathsf{T}$, where $\Gamma = \mathrm{diag}(\gamma_1, \ldots, \gamma_{M_s})$ and $H = \mathrm{diag}(\eta_1, \ldots, \eta_{M_t})$, the variables $\widehat{X} = U_s^\mathsf{T} X U_t$ and $\widehat{Q} = U_s^\mathsf{T} \widetilde{Q} U_t$ satisfy $\widehat{X} + \Gamma \widehat{X} H = \widehat{Q}$, so

$$\widehat{X}_{ij} = \frac{\widehat{Q}_{ij}}{1 + \gamma_i \eta_j}. \tag{60}$$

The solution is $X = U_s \widehat{X} U_t^\mathsf{T}$ and $V = K_s^{1/2} X K_{t,k}^{1/2}$.

Schur-complement elimination. Write $m_k = (m_{\beta,k}^\mathsf{T}, m_{u,k}^\mathsf{T})^\mathsf{T}$, so that $\Lambda_k m_k = \left((S_{\beta,0}^{-1} m_{\beta,0} + r_{\beta,k})^\mathsf{T}, r_{u,k}^\mathsf{T}\right)^\mathsf{T}$. Eliminating $m_{u,k}$ gives

$$\begin{aligned} P_{\beta,k} &= S_{\beta,0}^{-1} + R_{\beta\beta,k} - R_{\beta u,k} D_{u,k}^{-1} R_{u\beta,k}, \\ m_{\beta,k} &= P_{\beta,k}^{-1} \left(S_{\beta,0}^{-1} m_{\beta,0} + r_{\beta,k} - R_{\beta u,k} D_{u,k}^{-1} r_{u,k} \right), \\ m_{u,k} &= D_{u,k}^{-1} \left(r_{u,k} - R_{u\beta,k} m_{\beta,k} \right). \end{aligned} \tag{61}$$

Each product with $D_{u,k}^{-1}$ is computed by the Sylvester solve above.

B.7 Prediction

For a test input x_*, let $k_{u*,k} = \mathrm{Cov}(u_k, f(x_*))$, $a_{*,k} = K_{uu,k}^{-1} k_{u*,k}$, $\nu_{*,k} = k(x_*, x_*) - a_{*,k}^\mathsf{T} K_{uu,k} a_{*,k}$, and $h_{*,k} = (\phi(x_*)^\mathsf{T}, a_{*,k}^\mathsf{T})^\mathsf{T}$. Under $q_k(\beta, u_k) = \mathcal{N}(m_k, S_k)$,

$$\mathbb{E}_{q_k}[y_*] = \phi(x_*)^\mathsf{T} m_{\beta,k} + a_{*,k}^\mathsf{T} m_{u,k}, \qquad \mathrm{Var}_{q_k}(y_*) = \sigma_y^2 + \nu_{*,k} + h_{*,k}^\mathsf{T} S_k h_{*,k}. \tag{62}$$

The block inverse of Eq. (56) gives $S_{\beta\beta,k} = P_{\beta,k}^{-1}$, $S_{\beta u,k} = -P_{\beta,k}^{-1} R_{\beta u,k} D_{u,k}^{-1}$, and $S_{uu,k} = D_{u,k}^{-1} + D_{u,k}^{-1} R_{u\beta,k} P_{\beta,k}^{-1} R_{\beta u,k} D_{u,k}^{-1}$. With $v_{*,k} = D_{u,k}^{-1} a_{*,k}$,

$$h_{*,k}^\mathsf{T} S_k h_{*,k} = a_{*,k}^\mathsf{T} v_{*,k} + \left(\phi(x_*) - R_{\beta u,k} v_{*,k}\right)^\mathsf{T} P_{\beta,k}^{-1} \left(\phi(x_*) - R_{\beta u,k} v_{*,k}\right). \tag{63}$$

Prediction thus uses the same Sylvester solves as the posterior mean, without forming S_k.

B.8 Computational complexity

Let d_β be the trend dimension. The posterior computation requires eigendecompositions of an $M_s \times M_s$ spatial factor and an $M_t \times M_t$ temporal factor, together with matrix transforms and d_β Schur-complement solves. Its time complexity is

$$\mathcal{O}\big(M_s^3 + M_t^3 + (d_\beta + 1)(M_s^2 M_t + M_s M_t^2) + d_\beta^2 M_s M_t + d_\beta^3\big)\,. \tag{64}$$

Storing (R_k, r_k) requires the spatial and temporal factors, the off-diagonal block $R_{\beta u,k}$, the vector $r_{u,k}$, and the trend natural parameters, which take

$$\mathcal{O}\big(M_s^2 + M_t^2 + (d_\beta + 1)M_s M_t + d_\beta^2\big) \tag{65}$$

memory, where the extra $M_s M_t$ term accounts for $r_{u,k}$. For $M_s = M_t = M$ and fixed d_β, Eqs. (64)–(65) reduce to $\mathcal{O}(M^3)$ time and $\mathcal{O}(M^2)$ memory. In contrast, a direct solve with the full $M_s M_t \times M_s M_t$ matrix requires $\mathcal{O}(M_s^3 M_t^3)$ time and $\mathcal{O}(M_s^2 M_t^2)$ memory, that is, $\mathcal{O}(M^6)$ and $\mathcal{O}(M^4)$ in the balanced case.

This reduction is exact. The only approximations are the finite number of inducing variables, the random Fourier feature approximation of the temporal covariances (Section 3.3), and the conditional expectation in Section B.5. The Kronecker structure additionally assumes a separable kernel, fixed spatial inducing points, and every location observed at each time step.

B.9 Miller's backward recurrence

Eq. (31) requires the spherical Bessel values $j_m(x_j)$ for $m = 0, \ldots, M_t - 1$ and $j = 1, \ldots, D/2$, where $x_j = \lambda_j(t_k - t_0)/2$. Below, x denotes any one of the x_j. Forward recurrence is unstable when the order m is large relative to $|x|$, so we use Miller's backward recurrence (Gautschi, 1967, Sections 3 and 5).

We evaluate the recurrence at $|x|$ and use $j_m(-x) = (-1)^m j_m(x)$ for negative arguments. For the frequencies evaluated together, the starting order is

$$L = \left\lfloor M_t - 1 + \max_j |x_j| + 50 \right\rfloor. \tag{66}$$

Starting from $\widetilde{j}_L = 1$ and $\widetilde{j}_{L+1} = 0$, we compute

$$\widetilde{j}_{m-1}(x) = \frac{2m+1}{x}\widetilde{j}_m(x) - \widetilde{j}_{m+1}(x), \qquad m = L, L-1, \ldots, 1, \tag{67}$$

and normalize with $j_0(x) = \sin x / x$:

$$j_m(x) \approx \widetilde{j}_m(x)\,\frac{\sin x}{x\,\widetilde{j}_0(x)}, \qquad m = 0, \ldots, M_t - 1. \tag{68}$$

We use the fixed L in Eq. (66) without a convergence check.

To avoid overflow, the current iterates are rescaled when their largest absolute value exceeds 10^{100} or falls below 10^{-100} (10^{15} and 10^{-15} in single precision), and the accumulated scale factors are applied before normalization. The log-scale differences are clipped to $[-700, 700]$ ($[-80, 80]$ in single precision), and the smallest positive normal floating-point number is added to the denominator of Eq. (68). For $|x| \le 10^{-8}$, we set $j_0(x) = 1$ and $j_m(x) = 0$ for $m > 0$.

One pass costs $O(L) = O(M_t + \max_j |x_j|)$ operations per frequency, which is $O(M_t)$ when $|x|$ is bounded. For PEMS-BAY, j_m is computed with SciPy's `spherical_jn` instead.

C EXPERIMENTAL DETAILS AND ADDITIONAL RESULTS

C.1 Metrics and aggregation

For N held-out predictions with Gaussian posterior $y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, define $z_i = (y_i - \mu_i)/\sigma_i$. We report

$$\mathrm{RMSE} = \left[\frac{1}{N}\sum_{i=1}^{N}(y_i - \mu_i)^2\right]^{1/2}, \tag{69}$$

$$\mathrm{NLPD} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{2}\log(2\pi\sigma_i^2) + \frac{(y_i - \mu_i)^2}{2\sigma_i^2}\right], \tag{70}$$

$$\mathrm{CRPS}_i = \sigma_i\left[z_i\{2\Phi(z_i) - 1\} + 2\phi(z_i) - \frac{1}{\sqrt{\pi}}\right], \tag{71}$$

$$\mathrm{CRPS} = \frac{1}{N}\sum_{i=1}^{N}\mathrm{CRPS}_i, \tag{72}$$

where Φ and ϕ denote the standard-normal distribution and density. Calibration is measured over the central predictive intervals $\mathcal{A} = \{0.05, 0.15, \ldots, 0.95\}$:

$$\widehat{C}(\alpha) = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\left(|y_i - \mu_i| \le \Phi^{-1}\left(\frac{1+\alpha}{2}\right)\sigma_i\right), \tag{73}$$

$$\mathrm{ECE} = \frac{1}{|\mathcal{A}|}\sum_{\alpha\in\mathcal{A}}|\widehat{C}(\alpha) - \alpha|. \tag{74}$$

Coverage90 is $\widehat{C}(0.90)$ and is compared with its nominal value 0.90. RMSE, CRPS, NLPD, and ECE are lower-is-better. Tables report the mean and sample standard deviation across spatial splits. For sample-based predictive distributions, the same central-interval definition is evaluated using empirical predictive quantiles.

C.2 Evaluation protocol and baselines

We evaluate KronHiPPO-STGP under an online prediction protocol in which observations at held-out locations are unavailable at prediction time. Observations from visible locations are incorporated before prediction, and labels revealed after prediction at time t become available only for subsequent updates at $t + 1$.

All datasets follow the same causal evaluation principle: the predictive distribution at each step is computed using only information available before the prediction time, while newly revealed observations are used only in future posterior updates. Spatial splits are fixed for all methods within each dataset.

The comparison includes sparse and streaming variational GP baselines (OSGPR and StreamingSGPR), the HiPPO-based OHSVGP method, structured spatiotemporal and multi-output GP baselines (ST-SVGP, LMC-SVGP, ICM-SVGP, and FSDE-SVI), and IGNNK as a spatiotemporal kriging baseline. Each GP baseline is evaluated using its corresponding online update procedure, while IGNNK uses visible sensor inputs and the fixed road graph.

All GP baselines are calibrated on the initial task when applicable and then evaluated under their respective streaming procedures. We report the same spatial splits and label-availability boundary for all compared methods.

C.3 Dataset-specific configurations

ERA5-Land. The processed dataset contains 2-m dew-point temperature and associated meteorological covariates from the ERA5-Land hourly product (Copernicus Climate Change Service, 2019; Muñoz-Sabater et al., 2021) at 1,000 UK locations. Task 1 is used for empirical-Bayes calibration, feature scaling, and inducing-location selection. After calibration, (R, r) is reset to zero and updated through Tasks 2–10. The evaluation period contains 1,674 hours and is processed in 171 chronological batches. Five spatial splits are used, with 800 visible and 200 held-out locations in each split. The mean design contains seven deterministic time-coordinate features and 126

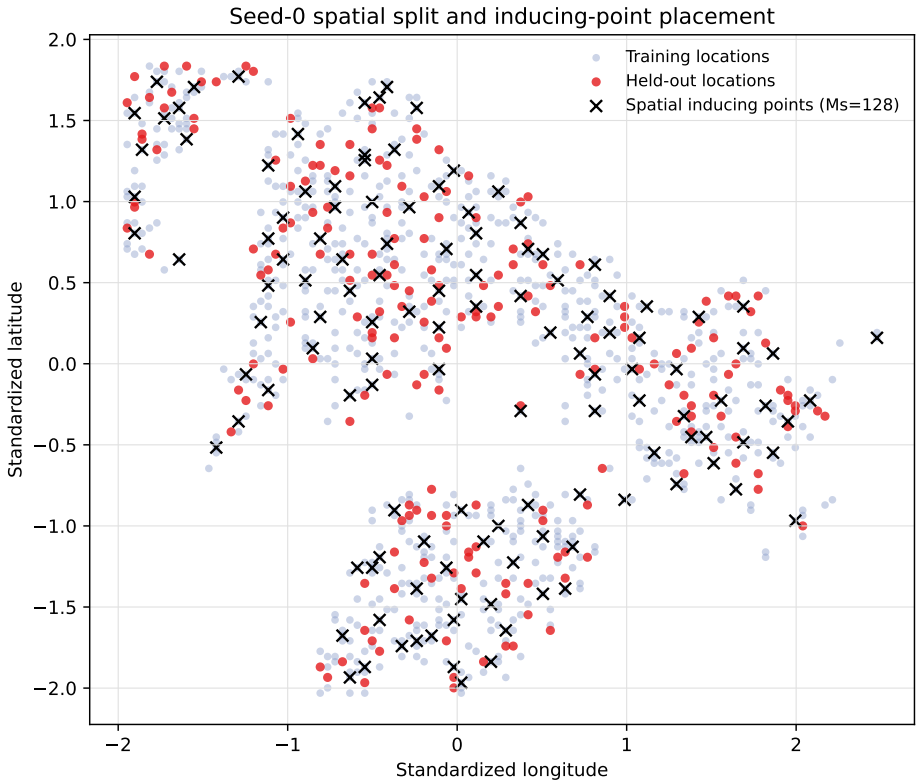


Figure 4: ERA5-Land spatial split and inducing-point placement. Visible locations, held-out locations, and spatial inducing points are shown for one representative split.

current, lagged, and difference features from six non-target meteorological variables; target lags are excluded. The benchmark configuration uses the default temporal kernel, at most 250 calibration iterations, validation every five iterations, and patience eight. The ablation configuration uses a three-component spectral-mixture temporal kernel with 100 fixed calibration iterations. Both use $M_s = M_t = 128$ and 256 random Fourier features. Figure 4 shows the spatial split and inducing points.

COVID-19. The COVID-19 data set contains 52 weekly jurisdiction-level hospital-admission series from the CDC NHSN Hospital Respiratory Data release (Centers for Disease Control and Prevention, nd). The response is the standardized $\log(1 + x)$ admission rate per 100,000 population. Kernel and likelihood hyperparameters are calibrated on Task 1 and fixed during the evaluation period. Held-out labels are not used to construct lag features.

PEMS-BAY. PEMS-BAY contains five-minute traffic-speed measurements from 325 Bay Area sensors in the public DCRNN benchmark release (Li et al., 2018). Task 1 initializes the posterior, and evaluation is performed over the subsequent 50,100 timestamps with online updates at every timestamp. Road-context lag features are constructed only from visible sensors. The model uses $M_s = 32$, $M_t = 128$, 512 random Fourier features, and the fixed road-context mean design. For illustration, Figure 2 shows sensors 400965 and 400178 in split 1, selected near the 50th and 85th percentiles of KronHiPPO-STGP RMSE.

Licenses. ERA5-Land is distributed under the CC BY 4.0 license, the CDC NHSN data are U.S. government works in the public domain, and PEMS-BAY is distributed with the DCRNN repository under the MIT license; the underlying data are from Caltrans PeMS. The solar irradiance data are distributed by NOAA NCEI under CC0, and the COVID-19 mortality data are derived from the SIVEP-Gripe database released by the Brazilian Ministry of Health under CC BY.

C.4 Full results

Table 4 gives the numerical values for Figure 1.

Table 4: Online prediction at held-out locations (mean ± SD; five splits for ERA5-Land/COVID-19 and three for PEMS-BAY). Bold and underlined values denote the best and second-best results; '–' indicates unavailable metrics.

Method	RMSE ↓	CRPS ↓	NLPD ↓	ECE ↓
ERA5-Land				
OSGPR (Bui et al., 2017)	0.9833±0.0127	0.5612±0.0086	1.5847±0.0294	0.1131±0.0051
StreamingSGPR (Maddox et al., 2021)	1.0660±0.0196	0.6283±0.0144	1.7331±0.0466	0.1543±0.0080
OHSVGP (Chen et al., 2025)	0.7303±0.0302	0.4132±0.0170	1.2989±0.0686	0.1063±0.0091
Kron-STGP	0.4166±0.0046	0.2391±0.0016	0.6590±0.0051	0.1711±0.0018
KronHiPPO-STGP	**0.1416±0.0036**	**0.0766±0.0017**	**−0.3349±0.0434**	**0.0669±0.0067**
COVID-19				
Adaptive OSGPR (Bui et al., 2017)	0.1742±0.0290	0.0990±0.0167	−0.2603±0.1757	0.1267±0.0435
ST-SVGP (Hamelijnck et al., 2021)	0.2094±0.0136	0.1192±0.0067	−0.0840±0.0390	0.0900±0.0239
LMC-SVGP (Jeong and Kim, 2023)	0.3376±0.0529	0.1965±0.0281	1.0809±0.4905	0.1806±0.0140
ICM-SVGP (Jeong and Kim, 2023)	0.3435±0.0447	0.2002±0.0219	1.0005±0.3588	0.1797±0.0016
FSDE-SVI (Jeong and Kim, 2023)	0.4705±0.0478	0.2580±0.0158	0.6693±0.1044	**0.0238±0.0066**
Kron-STGP	0.1605±0.0107	0.0922±0.0055	−0.3029±0.0533	0.1427±0.0240
KronHiPPO-STGP	**0.1580±0.0120**	**0.0885±0.0058**	**−0.3621±0.0583**	0.1183±0.0343
PEMS-BAY				
IGNNK (Wu et al., 2021)	0.7968±0.1226	–	–	–
OHSVGP	0.8190±0.0681	0.4064±0.0204	1.2312±0.0760	0.1671±0.0124
OSGPR	0.8207±0.0685	0.4130±0.0211	1.2318±0.0652	0.1816±0.0083
Kron-STGP	0.8139±0.0695	0.4104±0.0220	1.2242±0.0664	0.1827±0.0096
KronHiPPO-STGP	**0.7666±0.0629**	**0.3774±0.0193**	**1.1883±0.1076**	**0.1317±0.0196**

C.5 Ablation studies

Off-diagonal blocks and temporal basis. Within each dataset, the full model and the ablations use the same spatial split, calibration, and numbers of inducing variables. The ablation "w/o $R_{\beta u}$" sets $R_{\beta u,k}$ and $R_{u\beta,k}$ to zero after the natural parameters of the previous batches and the current batch have been added; all other steps are unchanged. The full-horizon-basis ablation keeps the off-diagonal blocks but replaces the evolving HiPPO basis with a fixed temporal basis over the whole time horizon.

Removing the off-diagonal blocks substantially degrades performance across both datasets. On ERA5-Land, the RMSE increases from 0.1251 to 1.2898, while on PEMS-BAY it increases from 0.7666 to 0.9555 (Table 5). Since the removed blocks include contributions from both previous and current batches, this ablation removes the coupling between β and u entirely, not only the contribution of previous batches.

Table 5: Full ablation results on ERA5-Land and PEMS-BAY (mean ± sample SD over spatial splits). Bold values denote the best result within each dataset; for Coverage90, the best value is the one closest to the nominal level of 0.90.

Variant	RMSE ↓	CRPS ↓	NLPD ↓	ECE ↓	Coverage90
ERA5-Land (five splits)					
Full model	$\mathbf{0.1251 \pm 0.0034}$	$\mathbf{0.0665 \pm 0.0014}$	$\mathbf{-0.6837 \pm 0.0186}$	$\mathbf{0.0461 \pm 0.0029}$	$\mathbf{0.9134 \pm 0.0035}$
w/o $R_{\beta u}$	1.2898 ± 0.0120	0.9951 ± 0.0088	52.8139 ± 1.9682	0.6905 ± 0.0018	0.1148 ± 0.0019
w/ full-horizon basis	0.1950 ± 0.0030	0.1420 ± 0.0009	0.2580 ± 0.0043	0.1807 ± 0.0012	0.9933 ± 0.0006
PEMS-BAY (three splits)					
Full model	$\mathbf{0.7666 \pm 0.0629}$	$\mathbf{0.3774 \pm 0.0193}$	$\mathbf{1.1883 \pm 0.1076}$	0.1317 ± 0.0196	$\mathbf{0.9032 \pm 0.0204}$
w/o $R_{\beta u}$	0.9555 ± 0.0232	0.4856 ± 0.0123	1.5333 ± 0.1323	$\mathbf{0.0474 \pm 0.0132}$	0.8490 ± 0.0217
w/ full-horizon basis	0.8112 ± 0.0665	0.4092 ± 0.0208	1.2194 ± 0.0637	0.1760 ± 0.0101	0.9249 ± 0.0112

With and without projection. Because the HiPPO inducing variables change at each update (Appendix B.3), we compare the update on ERA5-Land with and without the projection in Appendix B.4. Without projection, the previous natural parameters are reused as they are. The reference is the variational posterior recomputed from all observations so far, using the current inducing variables.

We report the marginal predictive Gaussian KL $\mathrm{KL}(p_{\text{online}} \| p_{\text{reference}})$, averaged first over query locations within each batch, then over batches within each split, and finally across splits. The first batch is excluded because no previous natural parameters exist and the KL is exactly zero.

Table 6: With and without projection at $M_t = 128$ (mean $\pm$ sample SD over five spatial splits).

Metric	With projection	Without projection
Predictive KL ↓	$(1.26 \pm 0.06) \times 10^{-4}$	2.0713 ± 0.1158
RMSE ↓	0.0795 ± 0.0051	0.2651 ± 0.0024
CRPS ↓	0.0463 ± 0.0017	0.1520 ± 0.0015
NLPD ↓	-0.9687 ± 0.0251	1.0741 ± 0.0960
ECE ↓	0.1913 ± 0.0085	0.1604 ± 0.0055
Coverage90	0.9825 ± 0.0038	0.6320 ± 0.0074

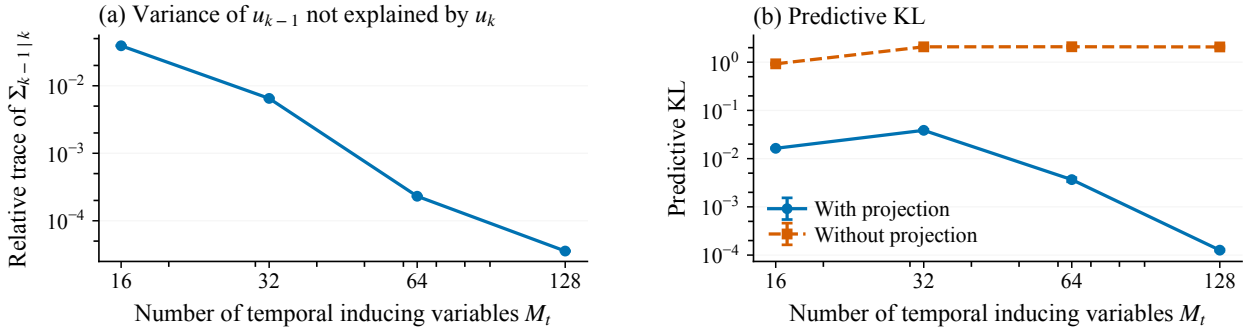


Figure 5: With and without projection. (a) $\mathrm{tr}(\Sigma_{k-1|k})/\mathrm{tr}(K_{uu,k-1})$, the fraction of the prior variance of u_{k-1} not explained by u_k. (b) Predictive KL to the reference posterior. With projection, the KL is not monotone in M_t; without projection, it stays at order one for $M_t \geq 32$.

C.6 Computational performance

The dense versus Kronecker-structured solver comparison was run on an NVIDIA GeForce RTX 5070 Laptop GPU under WSL; all other timing measurements were made on a single NVIDIA GeForce RTX 4090. To compare the cost of solving linear systems with $D_{u,k}$, the dense and Kronecker-structured implementations are given the same positive-definite precision matrices and right-hand sides.

Table 7 and Figure 3(b) report the results. Runtimes are means; speed-up is the mean of the within-replicate $t_{\text{dense}}/t_{\text{structured}}$ ratios and therefore need not equal the ratio of the mean runtimes. For $M_s = M_t = M$ and fixed trend dimension, the measurements are consistent with the $O(M^3)$ structured and $O(M^6)$ dense costs derived in Appendix B.8.

Table 7: Dense versus Kronecker-structured posterior solve. Relative error is $\|V - V_{\text{dense}}\|_F / \|V_{\text{dense}}\|_F$ for the solution V of Eq. (58) with a random right-hand side.

$M_s = M_t$	Dense (ms)	Structured (ms)	Speed-up ↑	Rel. error ↓
16	1.072	0.137	7.9×	9.2×10^{-14}
32	27.919	0.403	69.6×	9.1×10^{-13}
64	994.822	0.974	1036.7×	4.7×10^{-12}

Implementation benchmark. Table 8 compares the training time and peak GPU memory of KronHiPPO-STGP and ST-SVGP on ERA5 at $M_s = M_t = 128$.

Table 8: ERA5 implementation benchmark at $M_s = M_t = 128$ on an RTX 4090. The last row reports relative changes between KronHiPPO-STGP and ST-SVGP.

Method	RMSE	Train (s)	Profiler (GFLOPs/step)	Peak (GiB)
ST-SVGP	0.0882	4586.83	1106.04	14.75
KronHiPPO-STGP	**0.0837**	**41.39**	**27.36**	**1.68**
Relative comparison	5.1% lower	110.8× faster	40.4× lower	8.8× lower

Online efficiency. Table 9 reports the total runtime, update time per batch, prediction time, GPU memory, and memory for the stored statistics over ERA5 Tasks 2–10.

Table 9: Online efficiency summary for ERA5 Tasks 2–10. End-to-end timing includes all online batches.

Method	End-to-end (s)	Update/batch (s)	Predict (s)	GPU MiB	State MiB
OSGPR	78.00	0.1923	20.270	657	0.25
StreamingSGPR	29.81	0.0336	1.414	816	0.25
OHSVGP	254.32	1.2312	12.398	1966	0.08
KronHiPPO-STGP	79.52	0.0214	3.562	1067	19.16
Kron-STGP	61.06	0.0204	3.500	1069	20.92

C.7 Qualitative predictions

Figure 6 shows the ERA5-Land predictions at the held-out locations at the final batch.

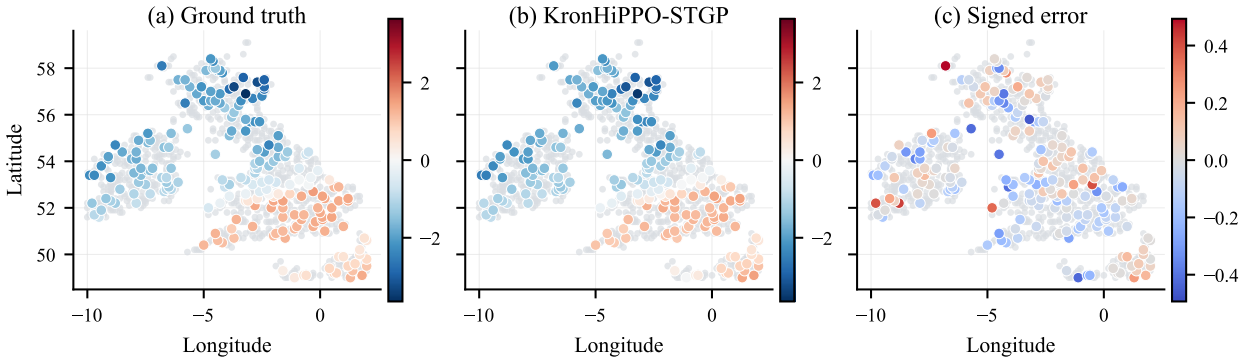


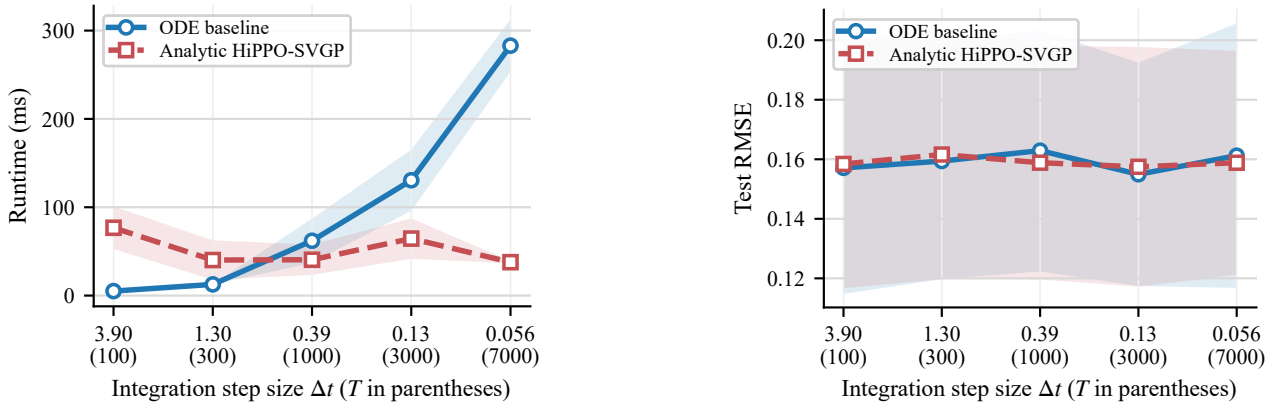
Figure 6: ERA5-Land predictions at the final batch of Tasks 1–10 for split 0. The panels show the standardized target field, the KronHiPPO-STGP posterior mean, and the prediction error (posterior mean minus target) at the 200 held-out UK locations; the remaining 800 visible locations are shown in light gray for spatial context.

C.8 Additional experimental details of analytic HiPPO covariance computation

We compare the analytic construction of $Z(t)$ in Section 3.3 with the ODE-based construction of Chen et al. (2025). To compare only the construction of $Z(t)$, these experiments use the temporal model of OHSVGP (Chen et al., 2025) rather than KronHiPPO-STGP; the two variants differ only in how $Z(t)$ is computed. We use two streams: solar irradiance (Lean, 2004) with 10 sequential tasks and a Gaussian likelihood, and daily COVID-19 mortality in Santa Catarina, Brazil (Hawryluk et al., 2021) with 5 sequential tasks and a Negative Binomial likelihood. The latter is a different dataset from the COVID-19 hospital-admission data in Section 4. Throughout this section, M denotes the number of HiPPO basis functions, which corresponds to M_t in the main text.

Runtime scaling of analytic HiPPO covariance computation. Here T denotes the number of ODE integration steps over a fixed interval, so the step size is $\Delta t = (t_{\max} - t_{\min})/T$; T is not the number of observations. Runtime is the wall-clock time of constructing the interdomain kernel matrices, measured as the median over 20 repeats after 5 warm-up runs on an RTX 4090 in double precision with 1000 random features. FLOP counts are estimated with the PyTorch profiler using 512 random features.

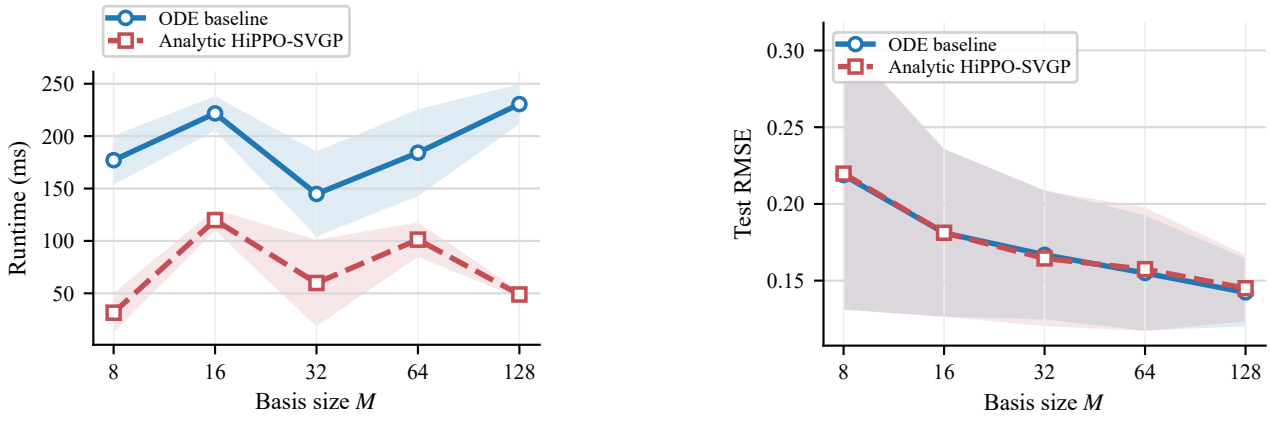
Figure 7 varies the integration step size at fixed basis size $M = 64$ on the solar data. As Δt decreases, the ODE construction must traverse more time steps, whereas the analytic construction evaluates the features directly at the target horizon. At the finest discretization, the analytic kernel construction is 7.5× faster, while the RMSE curves remain nearly overlapping. The FLOP estimates show the same trend: the ODE path grows from 1.02 to 59.37 GFLOPs, whereas the analytic path remains close to 0.177 GFLOPs.



(a) Kernel construction *(b) Test RMSE*

Figure 7: Kernel runtime and predictive RMSE versus integration step size Δt at fixed $M = 64$ on the solar data. Runtime lines show medians over 20 repeats, and shaded regions show one standard deviation.

Figure 8 varies the basis size at fixed discretization $T = 3000$. In this finer-discretization regime, the analytic construction is faster across all tested basis sizes, while both methods give similar test RMSE. At $M = 128$, the analytic kernel requires 0.236 GFLOPs compared with 101.30 GFLOPs for the ODE construction.



(a) Kernel construction *(b) Test RMSE*

Figure 8: Kernel runtime and predictive RMSE versus basis size M at fixed discretization $T = 3000$ on the solar data.

Solar experimental results. Table 10 reports the task-wise results, and Figure 9 shows the posteriors for selected tasks.

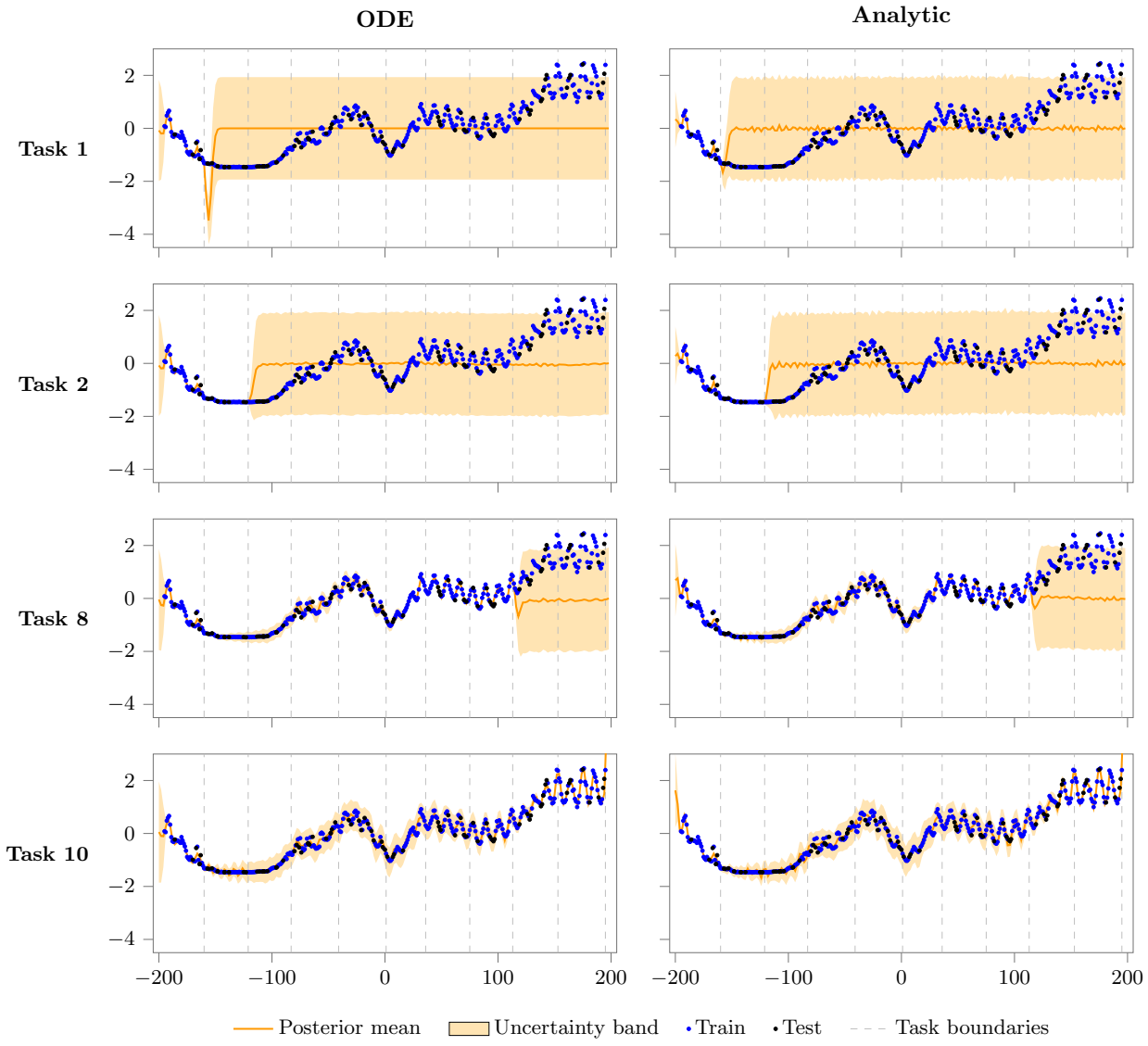


Figure 9: Solar posterior comparisons for the ODE-based and analytic HiPPO covariance constructions. Rows show Tasks 1, 2, 8, and 10 under the same train/test split; columns compare the ODE and analytic constructions. All panels use the same y-axis range. Task 2 gives the largest improvement of the analytic construction, Task 8 is the main counterexample, and Tasks 1 and 10 illustrate early- and late-stream behavior.

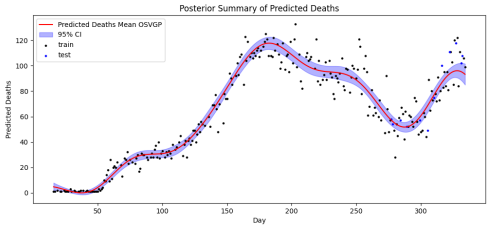
Table 10: Task-wise solar irradiance comparison between the ODE-based and analytic variants. The main Solar runs use basis size $M = 100$ and ODE step size $\Delta t = 2.45$. Negative deltas indicate an improvement of the analytic method over the ODE baseline.

Task	ODE RMSE	Analytic RMSE	Δ RMSE	ODE NLL	Analytic NLL	Δ NLL
1	0.1992	0.1802	-0.0190	8.0143	5.4882	-2.5261
2	0.1346	0.1146	-0.0201	4.5787	1.4713	-3.1074
3	0.1060	0.0911	-0.0149	1.9400	0.1451	-1.7949
4	0.1643	0.1621	-0.0021	7.8561	6.4982	-1.3578
5	0.1534	0.1509	-0.0025	6.5244	5.4849	-1.0395
6	0.1412	0.1457	0.0045	5.3541	2.5755	-2.7786
7	0.1298	0.1449	0.0152	1.3365	0.8754	-0.4611
8	0.1173	0.1467	0.0294	0.8595	1.5299	0.6704
9	0.1429	0.1296	-0.0133	1.5256	0.0976	-1.4281
10	0.1598	0.1521	-0.0077	2.3635	0.6493	-1.7143
Mean	0.1448	0.1418	-0.0030	4.0353	2.4815	-1.5537

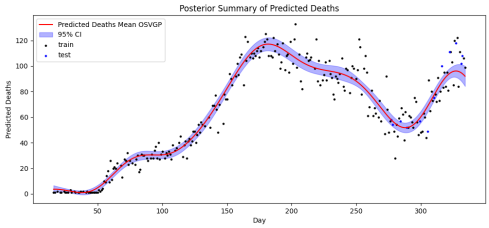
COVID-19 experimental results. Table 11 reports the task-wise results, and Figure 10 shows the posteriors for selected tasks.

Table 11: Task-wise COVID-19 mortality comparison between the ODE-based and analytic variants. The main COVID runs use basis size $M = 15$ and ODE step size $\Delta t = 2$. Each row evaluates on task "Eval" after training on tasks up to "After". Negative deltas indicate an improvement of the analytic method over the ODE baseline.

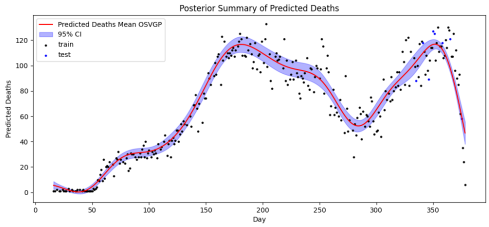
Eval	After	ODE NLPD	Analytic NLPD	Δ NLPD	ODE ECE	Analytic ECE	Δ ECE
1	1	2.9599	2.9636	0.0037	0.1500	0.1463	-0.0037
1	2	2.9783	2.9489	-0.0294	0.1278	0.1278	0.0000
2	2	3.7922	3.7910	-0.0011	0.0800	0.0717	-0.0083
1	3	2.9806	2.9527	-0.0279	0.1352	0.1352	0.0000
2	3	3.6662	3.6625	-0.0037	0.0767	0.0767	0.0000
3	3	3.8487	3.8628	0.0141	0.1183	0.1117	-0.0067
1	4	3.0697	2.9829	-0.0869	0.1907	0.1426	-0.0481
2	4	3.6496	3.6491	-0.0006	0.0767	0.0867	0.0100
3	4	3.8816	3.8965	0.0149	0.1283	0.1200	-0.0083
4	4	3.8058	3.8005	-0.0053	0.0517	0.0433	-0.0083
1	5	3.0870	3.0674	-0.0195	0.1537	0.1426	-0.0111
2	5	3.6310	3.6207	-0.0102	0.0867	0.1183	0.0317
3	5	3.8031	3.7754	-0.0277	0.0750	0.0767	0.0017
4	5	3.9312	3.9027	-0.0285	0.0533	0.0583	0.0050
5	5	4.1404	4.2054	0.0650	0.1375	0.1750	0.0375
Overall mean		3.5484	3.5388	-0.0095	0.1094	0.1089	-0.0006
Final mean		3.7185	3.7143	-0.0042	0.1012	0.1142	0.0129



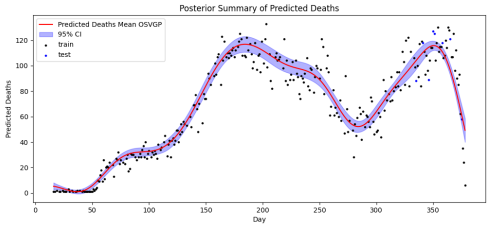
(a) Task 4, ODE



(b) Task 4, Analytic



(c) Task 5, ODE



(d) Task 5, Analytic

Figure 10: Representative COVID-19 posterior comparisons. Task 4 illustrates a case where the analytic construction slightly improves predictive density while largely preserving the posterior shape of the ODE baseline, whereas Task 5 shows the main final-stage counterexample where the analytic variant underperforms.